Do Fusion Operators Matter, and Does the Answer Survive Preprocessing? A Parameter-Matched Multiverse Comparison of Multimodal CNN Fusion for EEG Seizure Detection

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DRAFT v2, NOT FOR SUBMISSION. Restructured around the three research questions the advisor set on 10 October 2026. This file holds the Introduction (Section 1), the Methods (Section 2, in the advisor's subsections 2.1 to 2.7) and the Results of the five-seed runs (Section 3), the Discussion (Section 4) and the Conclusions (Section 5). The literature review is provisional: the systematic search has run, but title and abstract screening was done by an AI assistant, and the human screening and the full texts follow after 17 October; claims marked [abstract] rest on a source's abstract only and are to be checked against its full text. `paper/manuscript.md` is the previous draft (Bonn and CHB-MIT, single pipeline) and stays unchanged until this one replaces it; the Bonn benchmark moves to Supplementary Note 1 (advisor, 16 October 2026). Every number below is traceable to a file in the repository (Supplementary Table S11). Abstract and Introduction rewritten on the advisor's structure of 27 October 2026; the sources still to be checked against their full texts are listed in `docs/LITERATURE_CHECK.md`.

Abstract

A fusion operator that achieves the highest EEG seizure-detection score in one experiment may not remain superior when preprocessing, random seeds or evaluation conditions change. Yet architectural comparisons rarely test the robustness of such conclusions. We investigated whether fusion superiority survives these analytical choices. Four parameter-matched fusion architectures (early, late, gated and attention) and a separately evaluated score-fusion ensemble were compared with single-branch and classical baselines across nine preprocessing pipelines and five seed sets. Evaluation used subject-independent testing on CHB-MIT (23 subjects) and Siena (14 subjects), together with cross-dataset transfer. Statistical significance, practical equivalence and ranking stability were assessed separately. Median within-pipeline performance spreads were 0.021 and 0.053 macro F1 on CHB-MIT and Siena, respectively, with no operator pair significant in the primary corrected comparisons. Subject-associated variability accounted for approximately 74 percent of the estimated variance on CHB-MIT. Rankings varied across seeds and datasets; score fusion ranked first in 82 percent of Siena runs without establishing statistical superiority. Post hoc analyses showed that changing the decision threshold altered operator rankings, while prevalence-adjusted estimates on CHB-MIT gave lower macro F1 and a different preferred operator in more than half of the runs. Cross-machine differences in artefact removal also changed downstream detection outcomes. These findings distinguish high observed performance from robust comparative superiority and demonstrate the importance of evaluating EEG fusion under multiple analytical conditions rather than relying on a single favourable configuration.

1. Introduction

Automatic seizure detection from electroencephalography (EEG) has increasingly benefited from deep learning architectures capable of extracting complementary information from neural signals. Among these approaches, multimodal convolutional neural networks (CNNs) combine

time-domain EEG waveforms with time-frequency representations, typically through early fusion, late fusion, gating, attention or decision-level score aggregation. The underlying assumption is that integrating complementary representations can improve seizure discrimination beyond what either representation can achieve independently.

However, an important methodological question remains unresolved: when one fusion operator outperforms another, how much of that advantage can be attributed to the fusion mechanism itself, and how much depends on the analytical conditions under which the comparison was conducted? A model may achieve the highest performance in one experiment without retaining that advantage under alternative preprocessing pipelines, stochastic training runs, datasets or decision thresholds. Consequently, identifying the highest-scoring architecture and establishing robust architectural superiority are fundamentally different objectives.

1.1 Preprocessing as a source of analytical variability

EEG preprocessing is not merely a technical preparation step. Filtering, referencing, normalisation and artefact removal modify the signal characteristics available to downstream learning algorithms. These transformations may influence both predictive performance and the relative effectiveness of different fusion mechanisms.

Previous research has shown that deep-learning results can vary substantially across EEG preprocessing strategies. In the first systematic study of EEG preprocessing for deep learning, 4,800 models trained on six tasks with four architectures differed significantly between preprocessing pipelines; models trained on raw data ranked last on average, and minimal pipelines without artefact handling tended to serve them best (Del Pup et al. 2025) [abstract]. Artefact removal by independent component analysis, with Infomax or AMICA and three component rejection strategies, did not consistently improve deep decoding on three brain-computer interface datasets (Kang et al. 2024) [abstract], and even the scope of normalisation changed which strategy worked best (Truong and Delorme 2025) [abstract]. In none of these studies did a single approach consistently give the best outcome across tasks and models.

The broader multiverse-analysis literature provides a useful conceptual framework for understanding this problem. Rather than selecting one analytical pipeline and treating its outcome as definitive, a multiverse approach evaluates multiple defensible analytical choices and examines whether the conclusions remain stable (Steegen et al. 2016; Clayson 2024). In EEG research, no single pipeline was best on every quality criterion across 43 preprocessing pipelines (Huang et al. 2025) [abstract]; fourteen pipelines drawn from the N400 literature changed statistical outcomes, effect sizes and power (Šoškić et al. 2024) [abstract]; only 13 of 270 analyses of frontal alpha asymmetry in depression were significant (Kołodziej et al. 2021) [abstract]; and for event-related potentials, multiverse analyses have been used to choose pipelines by their effect on data quality and reliability (Clayson et al. 2021) [abstract].

For EEG fusion, this distinction is particularly relevant. A fusion operator may appear superior because its architecture is better suited to a specific preprocessing transformation rather than because it provides a generalisable advantage. A comparison performed under only one preprocessing configuration cannot distinguish these possibilities.

1.2 Architectural comparisons require controlled experimental conditions

Several EEG seizure-detection studies have compared fusion strategies or proposed increasingly sophisticated mechanisms for combining complementary representations (Table 1): five strategies within and across patients (An et al. 2026), four operators under leave-one-subject-out evaluation, where they lie within one standard deviation of each other (Einizade et al. 2023), three operators

with paired tests over six seeds (Basheer and Mishra 2026), and feature-level against decision-level fusion for seizure prediction (Daşdemir and Örnek 2025). These contributions have advanced multimodal EEG modelling, but their results also raise questions about the comparability of the evaluated architectures. Among the studies we have read in full, none matches the parameter counts of the compared operators, tests the differences between them with a correction for repeated resampling, and tests them for equivalence. Across more than 120 studies of hybrid EEG interfaces, only two explored more than one fusion strategy and one asked which was best (Lee et al. 2025).

Differences in parameter count, feature representation, optimisation settings, training duration and evaluation protocols may confound the interpretation of architectural comparisons. A more complex fusion model may outperform a simpler alternative because of its larger capacity or a more favourable training configuration rather than the fusion mechanism itself. Controlled comparisons in other domains point the same way: in ECG classification 1D and 2D representations were compared with early, late and joint fusion (Narotamo et al. 2024), a controlled benchmark on tabular neuroimaging data found the fusion strategy to matter more than the architecture (Camastra et al. 2026), and added multimodal complexity often did not pay off (Rheude et al. 2025, preprint).

Accordingly, comparisons designed to isolate the contribution of fusion should control the underlying backbone, training procedure and model capacity wherever feasible. Single-branch and classical machine-learning baselines are also important: an observed improvement over another fusion architecture does not necessarily demonstrate that multimodal fusion provides meaningful added value.

Even under controlled conditions, a higher mean performance score is not sufficient to establish superiority. Statistical uncertainty, the magnitude of the observed difference and the stability of the ranking must also be considered.

Table 1: Closest related work (from the systematic search of 3 October 2026; studies in bold are the closest to this one; the full texts of the remaining candidates are still being read).

Work	Contribution	What it leaves open
An et al. (2026), fastSeizureNet	Compares five fusion strategies for EEG seizure detection, within and across patients, with several backbones	Fuses hand-crafted with deep features rather than a waveform with its spectrogram; strategies not parameter matched and not tested against each other (a Wilcoxon test with Holm correction is used only against prior methods)
Einizade et al. (2023), fAttNet	Compares four fusion operators over raw EEG, wavelet packet and hand-engineered views, leave-one-subject-out on TUH	No parameter matching, no statistical test; the operators lie within one standard deviation of each other
Basheer and Mishra (2026)	Compares concatenation, element-wise addition and KAN gating, paired t tests over six seeds	Both streams come from the same raw EEG; uncorrected tests against concatenation only; no parameter matching
Daşdemir and Örnek (2025)	Protocol-controlled comparison of two fusion strategies for EEG	Seizure prediction, not detection; two operators; a 0.20 point accuracy difference without a stated test or parameter matching
Wang et al. (2026), *Neurocomputing*	Cross-attention fusion of time-domain and S-transform views, CHB-MIT and Siena	One ablation against concatenation, patient-specific evaluation, no statistical test
Chatzichristos et al. (2020)	Attention-gated multi-view fusion	One operator, no alternatives
Das et al. (2024); Huang et al. (2024)	1D+2D fusion; multi-head attention fusion	Fixed design, no operator ablation; gains without matched parameter counts

Work	Contribution	What it leaves open
Narotamo et al. (2024)	1D and 2D representations with early, late and joint fusion	ECG; no parameter matching or corrected and equivalence statistics
Camastra et al. (2026), *Brain Sciences*	Controlled fusion benchmark: strategy matters more than architecture	Tabular neuroimaging features; no paired or equivalence testing
Ali et al. (2024)	CHB-MIT results depend heavily on evaluation choices	Not about fusion operators
Dan et al. (2024), SzCORE	Standard datasets, event-based scoring and metrics for seizure detection	An evaluation framework
Kontras et al. (2026), NeuroAtlas	Bonn is saturated (eleven models at AUROC >= 0.99) and has no subject identifiers	Not a fusion comparison

1.3 Statistical superiority, practical equivalence and ranking stability are distinct

The reliability of model comparisons is further complicated by stochastic optimisation and dependence between evaluation folds. Different random seeds produce different fitted networks: even after weak models were excluded, accuracy differed by up to 10.8 percent between identical training runs, implementation-level factors alone caused differences of up to 2.9 percent, and only about one fifth of the surveyed papers used repeated runs to quantify this (Pham et al. 2020) [abstract]. Every source of nondeterminism in network optimisation, down to a one-bit change in the initial parameters, produces similar run-to-run variability, which makes small improvements hard to separate from noise (Summers and Dinneen 2021) [abstract], and seeds that perform far better or worse than average are easy to find (Picard 2021) [abstract]. In speech recognition, training variance was large enough to call for a change in how results are reported (van den Berg et al. 2017) [abstract], and seed-induced variability can be tested for directly (Banerjee et al. 2025) [abstract]. In repeated subject-independent evaluation, the training sets of different folds overlap, so conventional paired tests that treat the folds as independent are overconfident unless their variance is corrected (Nadeau and Bengio 2003; Bouckaert and Frank 2004), and whether a comparison comes out significant can depend on the number of folds and repeats (Jafrasteh et al. 2025).

Three complementary properties should therefore be distinguished. Statistical significance concerns whether the available evidence supports a difference between models under a specified inferential framework. Practical equivalence concerns whether the difference is sufficiently small to fall within a prespecified margin of practical relevance. Ranking stability concerns whether the relative order of models is preserved when analytical conditions change.

These properties are not interchangeable. Two models may show no statistically significant difference without being demonstrably equivalent: the absence of a significant difference is not evidence of equivalence. Conversely, one model may rank first consistently while its estimated advantage remains too uncertain to establish superiority. For fusion studies, evaluating these properties together provides a more informative assessment than selecting the architecture with the highest average score.

1.4 Dataset shift and evaluation choices challenge the generalisability of conclusions

Subject-independent evaluation is essential for investigating whether EEG seizure detectors generalise beyond the individuals represented in their training data. However, even leave-one-

subject-out evaluation within one dataset may not capture the variability introduced by different patient populations, acquisition systems, recording protocols and seizure distributions. High accuracies reported for automatic seizure detection often do not hold on new patients or recording sites. When 28 algorithms from the SzCORE challenge were evaluated on 4,360 hours of held-out continuous EEG from 65 subjects, the best F1 score was 32 percent, far below self-reported results, and the algorithms with the highest aggregate score were not the most consistent across subjects (Dan et al. 2025, preprint) [abstract]. Real-time detectors had been tested in distinct settings until they were compared under one (Lee et al. 2022, preprint) [abstract]; a prospective test on about 14,590 hours of adult EEG from another continent reported 76.68 percent with about 56 false alarms per 24 hours (Yang et al. 2021, preprint) [abstract]; and a systematic review names generalisability across training data and acquisition hardware among the main obstacles to clinical translation (Moutonnet et al. 2024, preprint) [abstract]. Within one public corpus, CHB-MIT results depend heavily on evaluation choices (Ali et al. 2024), and the Bonn corpus, long a standard benchmark, is saturated and carries no subject identifiers, so it cannot test generalisation across patients (Kontras et al. 2026).

Cross-dataset evaluation offers an additional test of generalisability. Importantly, transferring predictive performance does not necessarily imply that the relative ranking of fusion operators will also transfer. A model may retain acceptable aggregate performance while losing its position as the highest-ranked architecture.

Evaluation choices introduce another source of uncertainty. Window-level macro F1 depends on the threshold used to convert predicted probabilities into class labels, whereas event-level seizure sensitivity and false-alarm rates capture different aspects of detection performance; standard event-based scoring has been proposed to make such results comparable (Dan et al. 2024). Furthermore, artificially sampled class distributions can produce performance estimates that differ substantially from those expected under the much lower seizure prevalence of continuous EEG recordings, because precision depends on how many non-seizure windows each false-positive rate applies to.

Computational reproducibility is also relevant. CPU multithreading makes training nondeterministic (Xiao et al. 2021) [abstract], floating-point non-associativity in parallel computation causes run-to-run differences (Shanmugavelu et al. 2024) [abstract], reproducible training requires control of both software randomness and hardware nondeterminism (Chen et al. 2022) [abstract], and neuroimaging pipelines, including ICA decompositions of fMRI data, give different results on different operating systems (Glatard et al. 2015). Numerical differences introduced by hardware, software execution or artefact-removal procedures may therefore propagate to downstream predictions and potentially affect comparative conclusions.

Together, these considerations suggest that robust model selection requires evaluating not only whether a fusion operator performs well, but whether the evidence supporting its apparent advantage survives defensible changes in analytical and evaluation conditions.

1.5 Research gap and contributions

Despite growing interest in multimodal EEG seizure detection, an important gap remains between reporting the performance of competing fusion architectures and systematically evaluating the robustness of the conclusions drawn from those comparisons. The question this study asks is therefore: when a fusion operator appears superior, can that conclusion be trusted beyond the particular analytical conditions under which it was obtained?

The present study addresses this gap through a controlled, multiverse-inspired benchmark of fusion operators for EEG seizure detection. Four jointly trained fusion architectures, early, late, gated and attention fusion, are compared under parameter budgets matched within 0.4 percent, while

score fusion is evaluated as a separate decision-level ensemble outside the parameter-matched group. Single-branch networks and a classical machine-learning baseline provide additional controls.

The comparison is repeated across nine preprocessing pipelines, including three artefact removal algorithms, five seed sets and two independent EEG datasets, CHB-MIT and Siena, with leave-one-subject-out evaluation and cross-dataset transfer. Statistical significance, practical equivalence and ranking stability are examined separately. Additional reproducibility experiments investigate whether differences in artefact removal and computational execution can alter downstream conclusions.

Three research questions guide the investigation:

- **RQ1, performance robustness.** Do differences between fusion operators remain consistent across alternative EEG preprocessing pipelines and stochastic training conditions?
- **RQ2, ranking robustness.** Are conclusions about the highest-performing fusion operator stable across preprocessing strategies, random seeds, evaluation choices and computational environments?
- **RQ3, cross-dataset generalisability.** Do fusion performance patterns and operator rankings transfer between independent EEG seizure datasets?

The primary analyses answer these questions at the prespecified decision threshold, metrics and equivalence margins. Exploratory post hoc analyses, specified after the primary results were known, serve as robustness checks of the same questions: they re-examine the operator comparisons and rankings at other decision thresholds, at the seizure level, and under an adjusted class prevalence. They are not confirmatory tests.

The main contribution of this work is therefore not a new seizure-detection architecture, but a framework for assessing the robustness of comparative claims. By distinguishing observed performance, statistical evidence, practical equivalence and ranking stability, the study aims to establish when an apparent fusion advantage can be considered sufficiently reliable to support model selection. Along the way, the study documents corrections to the source data (a labelling error in CHB-MIT's annotations, two of its cases that are one person and must be held out together, a mains notch that has to follow each dataset's line frequency, and six problems in Siena's seizure lists), and releases the code, the per-fold results of every run and every analysis.

2. Methods

Figure 1 summarises the design: the datasets, the nine preprocessing pipelines, the two representations, the fusion operators and controls, and the evaluation; its panel b shows where each step is applied and how leakage between training and test data is prevented.

2.1 Datasets and EEG harmonisation

CHB-MIT (Shoeb 2009; Goldberger et al. 2000) holds 686 scalp recordings, 982.9 h, from 24 paediatric cases at 256 Hz in a bipolar montage. Two cases, chb01 and chb21, are documented as the same person and are treated as one subject, which gives 23 subjects. We use the standard 18-channel bipolar (double banana) montage and keep the 683 recordings that contain all 18 channels, with 185 of the 198 annotated seizures in the windowed corpus; the three dropped recordings belong to part of subject chb12's data recorded in a different montage. Seizure annotations were parsed by order rather than by their printed number, which corrects a labelling error in `chb09_08.edf`.

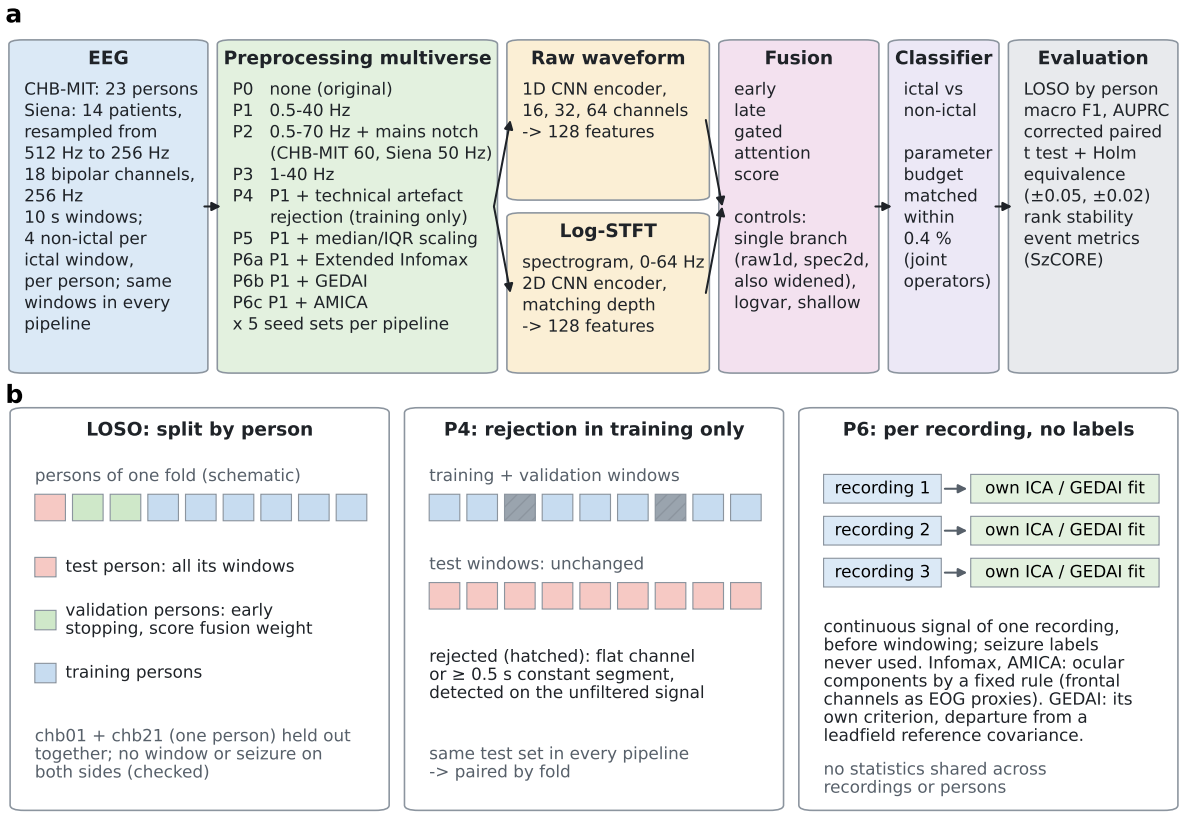


Figure 1: Study design. **a**, The multiverse: two datasets harmonised to 18 bipolar channels at 256 Hz and 10 s windows; nine preprocessing pipelines, each trained with five seed sets; a 1D CNN on the raw waveform and a 2D CNN on the log-STFT, combined by five fusion operators, four jointly trained under a matched parameter budget and score fusion as a decision-level ensemble of the two branches at their own size, with single-branch and classical controls; evaluation by leave-one-subject-out (LOSO) at the window and the seizure level. **b**, Where each step is applied and how leakage is prevented: whole persons are held out (chb01 and chb21 together), and validation persons serve only early stopping and the score fusion weight; P4 rejects windows from training and validation data only, so every pipeline is tested on the same windows; P6 cleans each recording on its own, without seizure labels.

Siena Scalp EEG (Detti et al. 2020) holds 41 recordings, 141.0 h by the EDF headers, from 14 adult patients, with 47 annotated seizures, recorded at 512 Hz in a referential 10-20 montage (older electrode names T3 to T6 are mapped to T7, T8, P7, P8). Six problems in the published seizure lists (a seizure ending after its recording, two conflicting onset or end times, two start times that disagree with the EDF headers, an onset given only clinically) were resolved before any analysis, each as either a correction verified against the recording or a stated decision rule; Supplementary Table S4 lists every decision, how it was checked, its remaining uncertainty, and how much the alternative would change the windows.

Harmonisation. Siena is converted to the CHB-MIT form before anything else: the 18 bipolar channels are formed as electrode differences of the referential signals, and the continuous recording is resampled from 512 to 256 Hz with a polyphase filter (`scipy.signal.resample_poly`, factor 1/2). Both corpora then pass through identical code. Seizure times are converted to seconds from each recording's first sample using the EDF header start time (modulo 24 h for recordings past midnight).

Windows and labels. Recordings are cut into non-overlapping 10 s windows. A window is ictal if it overlaps an annotated seizure at all. Non-ictal windows are subsampled per subject to four per ictal window of that subject, with a fixed seed, and every subject's true recording duration is kept for the event metrics (Section 2.5). This gives 6,380 windows (1,276 ictal) for CHB-MIT and 1,640 windows (328 ictal) for Siena, exactly four non-ictal windows per ictal window in every subject. The CHB-MIT windows come from 670 of the 683 recordings; no window was drawn from the other 13, all of them without seizures. The same windows are used by every pipeline, which is checked whenever a pipeline's corpus is built, so all comparisons between pipelines are paired window by window.

Class prevalence. The fixed ratio gives every pipeline, operator and subject the same class distribution, so that window-level results are comparable across them. Consequently, the window-level evaluation reflects this controlled sampling distribution, with 20 percent ictal windows, rather than the natural prevalence of seizures in continuous EEG, which is far lower (0.4 percent of all 10 s windows of the CHB-MIT recordings, 0.6 percent in Siena's recordings, which are cut around seizures). Window-level metrics, and macro F1 at a fixed threshold in particular, should therefore be interpreted within this experimental setting and not as the performance expected on continuous recordings. The seizure-level evaluation (Section 2.5) runs on the original, complete recording timelines and complements it.

Differences between the datasets. The two corpora differ in age (mostly children against adults), seizure types, recording length per patient (Siena's recordings are cut around seizures for a seizure prediction study), centre, equipment and mains frequency (60 Hz in Boston, 50 Hz in Italy). Results are compared between them as an independent replication and a transfer test (RQ3), not interpreted as an age or population effect: because the datasets differ in several of these respects at once, no difference between them can be attributed to any single one.

2.2 Preprocessing multiverse

Following the multiverse approach (Steegen et al. 2016), nine defensible preprocessing pipelines are crossed with every model, operator and seed instead of choosing one. Each pipeline changes one hypothesis-driven aspect; windows, labels, spectrogram settings, models, training and evaluation are fixed.

Pipeline	Band-pass	Notch	Artefact handling	Normalisation	Question
P0	none	none	none	z-score	original benchmark
P1	0.5 to 40 Hz	none	none	z-score	standard EEG band
P2	0.5 to 70 Hz	mains (CHB-MIT 60 Hz, Siena 50 Hz)	none	z-score	information above 40 Hz
P3	1 to 40 Hz	none	none	z-score	low frequency content and drift
P4	0.5 to 40 Hz	none	technical artefact rejection (training only)	z-score	artefact rejection
P5	0.5 to 40 Hz	none	none	median and IQR	normalisation
P6a	0.5 to 40 Hz	none	Extended Infomax ICA	z-score	artefact removal
P6b	0.5 to 40 Hz	none	GEDAI	z-score	artefact removal
P6c	0.5 to 40 Hz	none	AMICA	z-score	artefact removal

Order of operations. Each continuous recording is band-pass filtered, notch filtered (P2) and cleaned (P6), in that order; it is then cut into the fixed windows, P4's rejection rule is applied, and each window is normalised. The spectrogram is computed from the normalised window.

Filters. Zero-phase Butterworth filters (order 4, forward and backward) are applied to each continuous recording before it is windowed, so that no filter transient falls at a window edge. P2's notch removes each dataset's own mains frequency and its first harmonic; with a fixed 60 Hz notch Siena's P2 windows kept a 50 Hz line 318 times stronger than the 40 to 48 Hz background, so P2 was defined per dataset and its Siena runs were repeated.

Normalisation. Each channel of each window is standardised on its own: z-score (P0 to P4, P6) or median and interquartile range (P5). No statistic is computed across windows, recordings or persons, so normalisation cannot carry information from test to training data.

P4, technical artefact rejection. Amplitude was measured before a criterion was chosen: every amplitude threshold removes seizures selectively (500 μV rejects 72 percent of CHB-MIT's ictal and 48 percent of its non-ictal windows), because ictal windows have larger amplitudes. P4 therefore rejects only technical artefacts: a flat channel (standard deviation below 1 μV in the window) or a constant run of at least 0.5 s on any channel (a signal dropout), detected on the unfiltered signal. In CHB-MIT this rejects 42 of 1,276 ictal (3.3 percent) and 39 of 5,104 non-ictal windows (0.8 percent); the difference reflects which recordings have dropouts, not physiology (Supplementary Table S3). In Siena it rejects one window. Rejected windows are removed from training and validation only; the test windows are those of every other pipeline.

P6, three artefact removal algorithms. P6 applies Extended Infomax ICA (Lee et al. 1999; MNE-Python, Gramfort et al. 2013), GEDAI (Ros et al. 2025) or AMICA (Palmer et al. 2008; `jamica`) to the same 0.5 to 40 Hz signal, recording by recording. Infomax and AMICA share every setting and one component rejection rule: 16 components (the 18-channel bipolar montage has rank 16, because it contains two closed electrode loops per hemisphere), fitted on every 8th sample, with components removed when their correlation with any of the four frontopolar channels, used as EOG proxies since neither dataset has an EOG channel, has $|z| > 3$ (MNE `find_bads_eog`). Muscle components are not removed, because the gamma band carries the strongest ictal signal in CHB-MIT. GEDAI is not an ICA: it removes what departs from a leadfield-based reference covariance, by its own criterion, so the shared rule cannot apply. Its

reference covariance C, defined for a referential layout, is mapped onto the bipolar montage as $D\,C\,D^{\top}$ (D the electrode difference matrix), and the channel mean is removed before and restored after cleaning. GEDAI's threshold preset was fixed before any classification run, on signal preservation alone: on six CHB-MIT subjects the default removed more than half of the power of 39 percent of recordings, the conservative preset of 17 percent; the conservative preset removed no more from ictal than from non-ictal windows, whereas the default kept a lower median share of the ictal power (0.78 against 1.00; Supplementary Table S2); P6b uses the conservative preset. The comparison among P6a, P6b and P6c is therefore a comparison of whole cleaning procedures: for Infomax and AMICA it isolates the decomposition algorithm, while GEDAI differs also in how it decides what to remove. Signal retention and compute are in Supplementary Table S1, and the power removed by band and class in Supplementary Table S5.

No leakage through P6. Each recording is decomposed and cleaned on its own, from its own unlabelled signal; seizure annotations are never used, and no decomposition, threshold or statistic is shared between recordings or persons. A test person's recordings are therefore cleaned exactly as they would be without any training data.

Offline processing. Filtering and artefact cleaning are applied to complete recordings before windows are extracted, so a window's cleaned signal depends on data recorded after it. These experiments therefore evaluate the robustness of offline preprocessing and do not represent a causal, real-time seizure detection pipeline, in which only past signal is available.

2.3 EEG representations and fusion strategies

Representations. Each 10 s window enters the network twice: as the 18-channel waveform (2,560 samples per channel) and as its log-magnitude short-time Fourier transform, log(1 + |STFT|), computed per channel with a 1 s Hann window, 0.5 s hop and 256-point FFT and cropped at 64 Hz (65 frequency bins by 19 frames). The 64 Hz ceiling is kept in every pipeline, so input shapes and parameter counts never change.

Architecture. A 1D convolutional encoder (channel widths 16, 32, 64) processes the waveform and a 2D convolutional encoder of matching depth the spectrogram; each yields a 128-dimensional pooled feature vector. Five interchangeable fusion operators combine them:

- **Early:** the waveform passes a small strided 1D encoder, is broadcast along the spectrogram's time axis and concatenated to it as an extra channel before the 2D backbone (architecturally an intermediate fusion, since the two inputs share no tensor shape).
- **Late:** the two feature vectors are concatenated and classified by a small MLP.
- **Gated:** a learned scalar gate per branch reweights the concatenated features.
- **Attention:** a learned attention module weights the branches from both feature vectors jointly.
- **Score:** the two branches are trained as independent classifiers and their probabilities blended, p = w p_waveform + (1 - w) p_spectrogram, with the weight w chosen on the validation subjects only, by minimising their log loss over a 41-point grid from 0 to 1.

Parameter matching. Every jointly trained model (the four joint operators and two widened single-branch controls) is sized to a common budget by solving the fusion head's hidden width numerically: with 18 input channels the models have 102,236 to 103,712 parameters, the four joint operators within 0.36 percent of each other. Score fusion, an ensemble of two independently sized classifiers, keeps its natural size: it blends the probabilities of the two default-width single-branch controls below (28,850 and 34,546 parameters, 63,396 together, about 62 percent of the budget). Score fusion is retained as a complementary decision-level strategy: because its architecture and training differ from those of the jointly trained models, it is part of the broader practical

comparison of operators but not of the strictly parameter-matched architectural comparison, which concerns early, late, gated and attention fusion (102,547; 102,768; 102,921 and 102,767 parameters).

Controls. The two single-branch networks at their default width (raw1d, spec2d) and widened to the fusion budget (raw1d_wide, spec2d_wide), which asks whether fusion beats a unimodal model of the same size; and two classical baselines without representation learning, a logistic regression on log variance (logvar) and on seven hand-crafted features (log variance, line length and relative power in five bands; shallow), 19 and 127 parameters with the intercept. The networks see the same windows and preprocessing as the operators. The logistic regressions (standardised features, class-balanced weights, $C = 1$, no tuning) need no early stopping, so they are fitted on all training subjects, including the four (Siena: three) that the networks set aside for validation; this favours the classical baselines, if anything.

2.4 Experimental protocol

Leave one subject out. All evaluation is subject-independent. Each subject is held out once as the test set (23 folds on CHB-MIT, 14 on Siena), and all recordings of one individual are assigned to the same partition (chb01 and chb21 are held out together). Within each outer fold, validation subjects are selected exclusively from the remaining training subjects, and the held-out subject is excluded from all training, early stopping and model selection. The selection is deterministic and does not look at any result: the remaining subjects are ordered by their number of ictal windows, and round(0.2 n) of them (4 of 22 on CHB-MIT, 3 of 13 on Siena) are taken at evenly spaced positions of that order, so the validation set spans the range of seizure burdens and always contains ictal windows. It serves early stopping and the score fusion weight only. No subject's windows and no seizure's windows ever appear in two of the training, validation and test partitions; an automatic check of every fold confirms this for both datasets and that every stored run held out the same subjects in the same folds, and the leakage checker was itself validated on a deliberately leaky split (Supplementary Table S11).

Training. AdamW (Loshchilov and Hutter 2019), learning rate 1e-3, weight decay 1e-4, batch size 32, class-weighted cross-entropy; early stopping on validation macro F1 with patience 12 after a minimum of 20 epochs, at most 60 epochs. The minimum guards against a model stopping in a trivial single-class solution, found in an early audit. These settings, and the architectures, were fixed before the multiverse runs and not tuned: there was no hyperparameter search or optimisation, the same values were used for every model, pipeline and dataset, and the validation subjects served only early stopping and the score fusion weight. The reported performance is therefore that of a fixed, untuned configuration rather than the best each operator could reach.

Seeds. Every pipeline is run with five seed sets. A run's seed for each fold and model is derived from the seed set, fold and model, so it does not depend on execution order. It sets the weight initialisation, the order of the mini-batches and the dropout masks. It does not change the data: the window sampling (one fixed seed), the validation subjects (deterministic) and the artefact cleaning (a seed per recording) are the same in every seed set, so differences between seed sets reflect stochastic training alone. The ranking of the operators was unstable across three seed sets (Kendall's W below 0.5 in seven of nine CHB-MIT pipelines); after these three-seed results, the extension to five seed sets was decided, to estimate the seed variance more reliably. It was not a rule fixed beforehand.

Cross-dataset experiment (RQ3). Each model is trained once on all CHB-MIT subjects (with the same inner validation rule) and tested on every Siena window and recording, for every pipeline and seed set. The pipeline is applied to each dataset as defined, including each dataset's own mains notch in P2. Siena is used for testing only: the validation subjects are drawn from CHB-MIT, and no Siena window enters training, early stopping, the score fusion weight or

any other choice. Results are reported per Siena patient. The experiment asks whether the relative performance of the operators persists when the dataset changes. Because the datasets differ in population, acquisition and recording protocol (Section 2.1), it measures their combined distribution shift and does not isolate any single source of it.

2.5 Evaluation metrics

Window level (primary). Macro F1 and the area under the precision-recall curve (AUPRC) are primary; balanced accuracy (the mean of sensitivity and specificity) is reported alongside, with AUROC and the Brier score as secondary. A window is predicted ictal when its ictal probability exceeds 0.5, for every model and control alike. AUPRC is computed as average precision (scikit-learn) from the predicted ictal probability, with the ictal class as the positive class; it assesses how well the probabilities rank ictal above non-ictal windows across all thresholds, complementing the threshold-dependent macro F1. Metrics are computed per fold on the held-out subject's windows and averaged over folds within a run, each fold with equal weight. The subject is the unit of generalisation and of the statistical tests; pooling windows instead would let the subjects with the most seizures dominate the average.

Window-level classification and seizure-level detection are evaluated separately because they address different aspects of performance: macro F1 and AUPRC describe classification under the controlled window sampling (Section 2.1), whereas seizure-level sensitivity and false alarms per hour are assessed on the original recording timelines with event-scoring rules fixed in advance. A model can classify windows well and still miss seizures or raise many false alarms.

Seizure level (secondary, clinical). Every model saved for a fold is applied to every 10 s window of every recording of the held-out subject, not only the subsampled test windows, after the pipeline's own preprocessing; a window is positive when its probability exceeds 0.5, with no further post-processing. Detections are scored against the annotated seizures with SzCORE's event scoring (Dan et al. 2024; `timescoring` 0.0.7, defaults: 30 s tolerance before onset and 60 s after offset, any overlap counts, events closer than 90 s merged, events longer than 5 min split). Seizure-level sensitivity is the share of seizures detected; the false alarm rate is false-positive events per hour on the true recording time. The scored recordings hold 187 seizures in 979.9 h (CHB-MIT) and 47 seizures in 141.0 h (Siena). On CHB-MIT these are the 185 annotated seizures of the 683 recordings in the target montage (of 198 in the dataset, Section 2.1); one of them, in chb11_99, lasts 752 s and is split by the 5 min rule into three reference events, and in no other recording does merging or splitting change the count (Supplementary Table S11). Scoring the window labels themselves as predictions detects all of them without a false alarm, which checks the reference, the windows and the scoring together. On Siena only sensitivity is reported: its recordings are cut around seizures, so a false alarm rate would not describe continuous monitoring. The criterion, the merging of consecutive detections and the counting of false alarms were fixed, and committed, before any multiverse run, and are the same for every operator and pipeline.

2.6 Statistical analysis

Pairwise comparisons. Leave-one-subject-out folds share training data, so the standard paired test overstates significance; in a simulation of two equivalent classifiers the naive paired Wilcoxon test rejected in 37.6 percent of cases at a nominal 5 percent. On the Bonn corpus, studied in an earlier stage of this project, the naive test found 13 to 22 significant pairs among 55 per task and the corrected test none among the deep models (Supplementary Note 1). Because LOSO training sets overlap substantially, pairs of operators are compared with the resampling-corrected paired t test (Nadeau and Bengio 2003) rather than treating the held-out folds as independent

observations. The test inflates the variance of the mean fold difference from var(d)/k to (1/k + n_test/n_train) var(d), on k - 1 degrees of freedom. We count subjects: n_test/n_train = 1/22 on CHB-MIT and 1/13 on Siena, the held-out subject against the subjects outside it. Since 4 (Siena: 3) of these serve as validation subjects, the networks are fitted on 18 (10); with that conservative ratio, 1/18 and 1/10, no operator pair becomes significant and the pairs equivalent within 0.05 fall from 73 to 70 of 90 on CHB-MIT and from 21 to 19 on Siena. Subjects contribute different numbers of windows, but each fold enters the test as one observation (Section 2.5). Seed sets are averaged within each fold before testing, so the five seed sets and nine pipelines never inflate the number of independent observations, which is always the number of subjects.

Multiplicity is addressed with Holm correction within predefined families: the ten operator pairs of one pipeline, for one dataset and metric, form one family (nine families per dataset); the 30 comparisons of the six controls with the five operators in one pipeline form another.

Practical equivalence. For each pair the two one-sided test with the same corrected variance gives the smallest margin within which the pair is equivalent at alpha = 0.05; a pair is equivalent if that margin is at most 0.05 macro F1 (main analysis) and, as a stricter sensitivity analysis, 0.02. Both margins were fixed and committed before any multiverse result existed. A pair is robustly equivalent if it is equivalent in every pipeline. Statistical significance and practical equivalence are assessed separately: failure to reject equal performance is not interpreted as evidence of equivalence. The corrected test, the equivalence bound, the Holm adjustment and Kendall's W were checked against independent re-implementations (Supplementary Table S11).

Ranking stability. In every run (pipeline and seed set) the five operators are ranked by mean macro F1. We report each operator's full rank distribution (counts per rank, median, best and worst rank, share of runs won and lost), Kendall's W across seed sets within each pipeline, and Kendall's tau between the seed-averaged rankings of every pair of pipelines. Whether preprocessing changes the ranking more than the seed does is tested by comparing the mean tau between seed sets of the same pipeline with the mean tau between pipelines, against a null distribution from 10,000 permutations of the pipeline labels within each seed set; the same test is run among P6a, P6b and P6c alone.

Fusion x Preprocessing. A repeated-measures ANOVA on macro F1 (seed means) treats the held-out subjects as the repeated units and pipeline and operator as within-subject factors; each effect is tested against its interaction with subject, with Greenhouse-Geisser correction. Effect sizes are reported as partial eta squared.

Variance components. For macro F1, AUPRC and balanced accuracy, the fully crossed subject x pipeline x operator x seed design is decomposed as y = mu + subject + pipeline + operator + seed + every interaction, all treated as random effects (generalizability theory) and estimated by the method of moments from the expected mean squares. With one observation per cell the four-way interaction is the residual. Every component is reported unrounded, as an estimated variance, as a standard deviation in the metric's own units (a practical size) and as a share of the total; negative estimates, which arise when an effect's mean square is smaller than its error term, are reported as such and set to zero in the shares. The components are descriptive estimates of the variability associated with the held-out subject, the pipeline, the operator and stochastic training. A share of zero is therefore not proof that an effect is absent, and the subject component is read as subject-associated variability: it collects everything that differs between held-out subjects, of which individual physiology is only one part, together with seizure count and type, recording length and recording conditions.

Prediction stability. The agreement (Cohen's kappa) of window-level predictions between pipelines at the same seed set is compared with the agreement between seed sets in the same pipeline.

Statistical significance is reported from the corrected tests and the ANOVA, practical size from

the equivalence margins and the variance components in metric units, and ranking stability from the rank analyses; a result in one of these three is never used as evidence for another. All analyses were validated on synthetic runs with known effects (39 checks).

Post hoc sensitivity analyses. To examine the sensitivity of the main conclusions to evaluation choices, we conducted four exploratory post hoc analyses after completion of the primary multiverse experiments. First, the stored window-level predictions of every model, in all nine pipelines, five seed sets and both datasets, were re-evaluated at decision thresholds of 0.3 and 0.7, in addition to the prespecified threshold of 0.5. Operator rankings and paired comparisons were recalculated with the original subject-level evaluation units (the corrected test on fold values, seed sets averaged). Holm-adjusted comparisons were examined both within each threshold (the ten operator pairs of one dataset, pipeline and threshold, the family used above) and, as an additional multiplicity sensitivity check, across the three thresholds (the thirty comparisons of one dataset and pipeline). Second, seizure-level sensitivity and false-alarm rates were re-evaluated at the same thresholds for pipelines P0 and P1 (five seed sets each), applying the saved models to the complete recordings of both datasets without retraining or post-processing; false alarms per hour are reported for CHB-MIT only. Third, we estimated the effect of the 1:4 non-ictal-to-ictal sampling on macro F1 for CHB-MIT by reweighting each held-out subject's confusion matrix at the 0.5 threshold: every non-ictal test window was weighted by the ratio of its case's non-ictal windows in the complete recordings to those sampled (15 to 334), ictal windows by one, so that true positives and false negatives keep their counts while false positives and true negatives are scaled to the full non-ictal count. Macro F1 was computed per held-out subject and averaged with equal weight, as in the primary analysis; a pooled version, which sums the confusion matrices of all subjects first, is reported alongside (Supplementary Table S10b). Operator pairs were compared on the prevalence-adjusted fold values with the corrected test and the Holm families of the primary analysis (Supplementary Table S10d). These prevalence-adjusted estimates assume that the sampled non-ictal windows are representative of the non-ictal windows not included in the primary evaluation; they do not constitute direct evaluation on the complete natural-prevalence window distribution. Finally, subject-level paired bootstrap intervals were computed as an alternative assessment of inferential sensitivity: 10,000 resamples of the held-out subjects, in which both operators' results of a drawn subject enter together, with percentile 95 percent intervals, and a variant that also resamples the five seed sets within each drawn subject. Unlike the corrected test, the bootstrap does not model the overlap of the LOSO training sets: it treats the held-out subjects as independent draws. The bootstrap intervals and the corrected 95 percent intervals set beside them are nominal, not adjusted for multiplicity, so they are compared with each other and not with the Holm-adjusted primary results. Because these analyses were specified after inspection of the primary results, they were treated as exploratory robustness assessments rather than confirmatory hypothesis tests. The primary corrected comparisons and the prespecified equivalence analyses remained unchanged (Supplementary Tables S9 and S10).

2.7 Reproducibility and computational cost

Software and hardware. Two environments were used (Table 2); artefact removal needs a newer numpy than the training code and runs in its own environment (`requirements.txt`, `requirements-ica.txt`). The software versions and the hardware of every run are recorded in its `meta.json`.

Table 2: Computational environments.

	Training and window-level evaluation	Artefact removal (P6) and seizure-level evaluation
Used for	all LOSO and cross-dataset runs (135 runs)	the P6 corpora; applying the saved models to every window of every recording
Machine	lab PC: 2 x Intel Xeon (Skylake-SP), 64 logical CPUs; NVIDIA GeForce RTX 5070	laptop: Intel Core Ultra 7 155U, 14 logical CPUs; no CUDA GPU
Operating system	Windows 10 (build 19045)	Windows 11 (build 26300)
Python	3.12.10	3.12.7
Numerical libraries	PyTorch 2.8.0 (CUDA 12.8 build), numpy 1.26.4, scipy 1.16.1, scikit-learn 1.7.2	PyTorch 2.8.0 (CPU build), numpy 2.5.3 (OpenBLAS), scipy 1.16.1, scikit-learn 1.7.2, MNE 1.13.2, GEDAI 0.60, jamica 0.3.0, timescoring 0.0.7
Threads	16 CPU threads	14 (BLAS)
Seeds	per seed set, fold and model (`fold_seed`)	per recording, from its file name (crc32)
Determinism	PyTorch deterministic algorithms on the GPU	CPU only

Determinism and device. All training runs of one analysis use one machine, one device and 16 CPU threads, and CPU and GPU runs are never mixed in one analysis. On the CPU a run was reproduced bit for bit, also with two runs sharing the machine. The multiverse ran on the GPU with PyTorch's deterministic algorithms; adaptive average pooling has no deterministic CUDA backward, yet two GPU runs of the same fold were identical. Before switching to the GPU, its agreement with the CPU was tested at identical seeds: over five folds the mean absolute macro F1 difference between CPU and GPU was 0.074, slightly larger than the 0.066 between two CPU seed sets. The criterion fixed beforehand, a CPU-GPU difference no larger than the difference between seed sets, was therefore narrowly missed, though the CPU-GPU difference is still of the same order as the variation between stochastic training runs; this similarity does not establish that hardware-related numerical differences and seed effects arise from the same mechanism, and we make no claim about the cause.

Reproducibility conditions. Reproducibility was examined at the level of the preprocessing and of the downstream predictions (Table 3). Identical preprocessing configurations were run in different computational environments, and the differences in the cleaned signals were followed into the classification and seizure-level metrics.

Table 3: What was held fixed and what varied in each reproducibility comparison.

Comparison	Held fixed	Varied	Measured
Training, repeat on CPU	code, data, seeds, machine, threads	nothing (also two runs sharing the machine)	fold metrics: bit-identical
Training, repeat on GPU	code, data, seeds, machine, GPU, deterministic mode	nothing	fold metrics: identical
Training, CPU against GPU	code, data, seeds, machine	device (CPU or GPU)	macro F1 difference, against that between seed sets
Artefact removal, two machines	code, input recordings, package versions, seeds, BLAS kernel	machine (laptop, lab PC), thread count	components removed, power kept, per recording; 670 CHB-MIT recordings
Artefact removal, same machine	code, input, versions, seeds, machine	process parallelism (run alone or in parallel workers)	as above

Comparison	Held fixed	Varied	Measured
Artefact removal, third machine	code, input, versions, seeds	architecture (ARM, macOS)	as above, on a subset of recordings
Rounding perturbation	everything	input multiplied by 1 + 1e-7 N(0, 1)	as above
Effect on outcomes	trained models, test recordings	cleaned signals (training-time against event-time cleaning)	seizure-level sensitivity, false alarms, rankings

Reproducibility of artefact removal. Fixing seeds and package versions did not make P6 reproducible across machines and execution contexts. AMICA removed the same components in every recording on every machine and in every execution context; the power it kept differed by at most 0.003 between machines and 0.0007 between two runs on the same machine. Extended Infomax removed different components in recordings whose component lay near the $|z| = 3$ rule: 2.5 percent of recordings between two machines and 1.2 percent between two runs on the same machine with different process parallelism. GEDAI was deterministic on each machine but gave a different result on the second machine in 27 percent of recordings, often by a large amount. A third machine of another architecture (ARM) agreed with the other two on recordings where they agreed, and often with neither where they differed. Multiplying the input by (1 + 1e-7 N(0, 1)), a rounding-level perturbation, changed GEDAI's result on exactly the recordings that differed between machines, also on a full-rank montage, and Infomax's component decision on recordings where it had differed; AMICA was unchanged. This is consistent with a sensitivity of the methods to rounding near a decision threshold on particular recordings rather than with a fault of one machine or library, but it does not identify the numerical mechanism, and the differences are reported as an empirical reproducibility limitation. All three methods were compared on the same criteria: the components removed and the share of signal power kept, per recording. Window-level results are unaffected, because every P6 training and test window comes from one set of signals, cleaned once. The event metrics, which need every recording cleaned again, were computed on the machine that produced those signals; they then reproduce the training-time cleaning exactly for GEDAI, within 1.4 percent of the amplitude for AMICA, and differ for Infomax in 8 of 670 CHB-MIT recordings (`events/meta.json`, `window_check`).

Computational cost. Artefact removal of the 670 CHB-MIT recordings with selected windows took 4.7 (GEDAI), 8.5 (Infomax) and 73 (AMICA) CPU hours on a laptop CPU (Supplementary Table S1). One CHB-MIT run (23 folds, nine models) took about 1.5 h on the GPU, one Siena run about a quarter of an hour, and one cross-dataset run about 5 min; training one fold (all nine models) took 3.5 to 6 min on the GPU against 16 to 17 min on the CPU (`perfold.csv`).

Availability. Code, the per-fold results of every run, and all analyses are in the public repository, whose README lists the commands that regenerate each table and figure. Following our own recommendation (Section 4.5), the repository also holds, per recording and artefact removal method, the outcome of the cleaning the models were trained and tested on (components removed, GEDAI's threshold, power kept) and of the recleaning used for the event metrics, so that a reader can check a reproduction recording by recording. Supplementary Table S11 maps every analysis of the manuscript to the code that runs it and the files that hold its results.

3. Results

The evaluation comprised nine preprocessing pipelines, five seed sets and five fusion operators, together with single-branch neural-network and classical controls. Each pipeline-seed configuration was evaluated with leave-one-subject-out (LOSO) cross-validation on CHB-MIT (23 subjects)

and Siena (14 subjects), giving 45 runs per dataset; 45 further runs evaluated transfer from CHB-MIT to Siena. Means over runs are given with the standard deviation over runs.

The results are presented in eight stages. We first examine whether fusion operators offer a meaningful performance advantage and whether it justifies their architectural complexity (RQ1; Section 3.1). We then distinguish statistical significance from practical equivalence (RQ1; Section 3.2), assess the stability of the operators' ranking across pipelines and seeds (RQ2; Section 3.3), and identify the principal sources of performance variability (RQ1 and RQ2; Section 3.4). We then examine the effects and the reproducibility of artefact removal (RQ2; Section 3.5), the seizure-level detection (Section 3.6), and whether the performance patterns and rankings generalise across datasets (RQ3; Section 3.7). Finally, the exploratory post hoc analyses check the robustness of these answers to the decision threshold and the class prevalence (Section 3.8).

3.1 Fusion complexity does not consistently translate into superior performance

The five fusion operators reached broadly comparable window-level performance across the preprocessing multiverse, although their relative behaviour differed between datasets (Table 4, Figure 2).

On CHB-MIT, mean macro F1 over the 45 runs ranged from 0.673 ±0.017 for score fusion to 0.686 ±0.016 for gated fusion; early, late and attention fusion reached 0.677 ±0.013, 0.681 ±0.015 and 0.683 ±0.014. Within a pipeline, the best and the worst operator differed by 0.013 to 0.035 macro F1 (median 0.021), and each operator's mean varied little across pipelines (SD 0.007 to 0.008).

On Siena, the operators differed more. Score fusion reached the highest mean macro F1 (0.670 ±0.024), followed by late (0.640 ±0.025), gated (0.631 ±0.031), attention (0.625 ±0.023) and early fusion (0.617 ±0.021). Within a pipeline the best and the worst operator differed by 0.042 to 0.070 (median 0.053), and the variation across pipelines was larger than on CHB-MIT (SD 0.011 to 0.024). The separation between operators was therefore dataset-dependent, and its size has to be weighed against its uncertainty (Section 3.2) rather than read as superiority from mean performance alone.

Increasing architectural complexity did not consistently improve performance. The spectrogram network widened to the fusion budget reached a mean macro F1 of 0.683 on CHB-MIT, within the range of the jointly trained operators (0.677 to 0.686), and 0.663 on Siena, above every jointly trained operator (0.617 to 0.640) and below only score fusion (0.670); it exceeded the best operator of a run in 9 (CHB-MIT) and 16 (Siena) of 45 runs. The waveform network was weaker (0.662 and 0.612 at the fusion budget, 0.663 and 0.609 at default width), and widening either branch to the fusion budget changed little (Supplementary Table S7). The shallow classifier, a logistic regression with 127 parameters on seven hand-crafted features per channel, was equally competitive: its mean macro F1 was 0.690 on CHB-MIT, above the best operator of a run in 12 of 45 runs, and 0.720 on Siena, above the best operator in 43 of 45 runs and in all nine pipelines. This advantage depends on the decision threshold on CHB-MIT, where the shallow classifier's macro F1 is the lowest of all models at 0.3 (0.642) and the highest at 0.7 (0.715), but not on Siena, where it is the highest at 0.3, 0.5 and 0.7 (Supplementary Table S9a).

These descriptive advantages did not amount to statistically significant superiority. Tested like the operator pairs (corrected test, Holm across the 30 control-operator comparisons of a pipeline), no control differed significantly from any operator in any pipeline of either dataset, in macro F1 or AUPRC; the smallest corrected p was 0.12 (shallow against early fusion on Siena), and equivalence within 0.05 macro F1 could be shown in 34 of 270 comparisons on CHB-MIT and 13 of 270 on Siena (Figure 2c, f). The controls therefore stood to the operators as the operators stood to each other, and the additional complexity of multimodal fusion was not accompanied by a reliable improvement in macro F1 under the conditions evaluated.

Classification at a threshold and the ranking of windows by probability behaved differently. On CHB-MIT gated fusion had the highest mean AUPRC (0.666 ±0.018) and early fusion the lowest (0.613 ±0.024); on Siena early fusion reached 0.496 ±0.035 against 0.631 ±0.033 for score fusion. Early fusion had the lowest AUPRC in all nine pipelines of both datasets (0.564 to 0.632 on CHB-MIT against at least 0.631 for every other operator, 0.453 to 0.540 on Siena against at least 0.522), although its macro F1 at the 0.5 threshold differed much less. On CHB-MIT the fused models ranked windows better on average than a single branch (late, gated, attention and score fusion 0.655 to 0.666 against at most 0.618); on Siena the difference from the best single branch (0.581) was inconsistent, from -0.085 (early fusion) to +0.050 (score fusion). Score fusion, which blends the two default-width single-branch networks, had the lowest Brier score of all models (0.265 and 0.304), and early fusion the highest among the operators on CHB-MIT (0.330 against 0.294 to 0.300). Fusion complexity, thresholded classification and probability ranking are therefore distinct properties: an operator can reach a competitive macro F1 without discriminating equally well across thresholds.

Table 4: Window-level performance of the fusion operators and the main controls, mean and SD over the 45 runs of each dataset (fold means within a run). Per pipeline: Figure 2 and Supplementary Table S6; all six controls with their parameter counts: Supplementary Table S7.

Model	CHB-MIT macro F1	CHB-MIT AUPRC	CHB-MIT bal. acc.	Siena macro F1	Siena AUPRC	Siena bal. acc.
early	0.677 ±0.013	0.613 ±0.024	0.682 ±0.011	0.617 ±0.021	0.496 ±0.035	0.642 ±0.022
late	0.681 ±0.015	0.656 ±0.019	0.679 ±0.012	0.640 ±0.025	0.599 ±0.040	0.679 ±0.018
gated	0.686 ±0.016	0.666 ±0.018	0.683 ±0.013	0.631 ±0.031	0.591 ±0.034	0.674 ±0.022
attention	0.683 ±0.014	0.655 ±0.020	0.682 ±0.013	0.625 ±0.023	0.580 ±0.034	0.667 ±0.020
score	0.673 ±0.017	0.655 ±0.017	0.678 ±0.015	0.670 ±0.024	0.631 ±0.033	0.689 ±0.022
raw1d_wide (control)	0.662 ±0.012	0.605 ±0.017	0.673 ±0.010	0.612 ±0.028	0.577 ±0.033	0.657 ±0.019
spec2d_wide (control)	0.683 ±0.014	0.618 ±0.016	0.680 ±0.013	0.663 ±0.026	0.581 ±0.027	0.681 ±0.021
shallow (control)	0.690 ±0.008	0.640 ±0.008	0.739 ±0.007	0.720 ±0.019	0.667 ±0.015	0.770 ±0.017

3.2 Small performance differences do not establish practical equivalence

The narrow differences between operators called for a formal distinction between statistical significance and practical equivalence (Figure 3).

In both datasets, none of the 90 pairwise operator comparisons (10 pairs in each of nine pipelines) was significant after Holm correction; the smallest corrected p was 0.19 on CHB-MIT and 0.095 on Siena. The absence of a significant difference, however, did not mean that the operators were practically equivalent.

At the margin of ±0.05 macro F1 fixed in advance, 73 of the 90 comparisons on CHB-MIT were equivalent, and two pairs, late-attention and gated-attention, were equivalent in all nine pipelines. At the stricter margin of ±0.02 only one comparison was. On Siena the evidence for equivalence was much weaker: 21 of 90 comparisons were equivalent within ±0.05, no pair in every pipeline, and none within ±0.02. The smallest margin at which equivalence could be shown had a median of 0.040 on CHB-MIT (0.020 to 0.066) and 0.081 on Siena (0.021 to 0.148).

The two datasets therefore supported different inferences. CHB-MIT showed that many operators differ by less than a moderate margin, whereas Siena often lacked the precision to establish either a difference or equivalence. The results thus fall into three kinds: statistically detectable differences (none here), demonstrated practical equivalence, and inconclusive comparisons; treating every

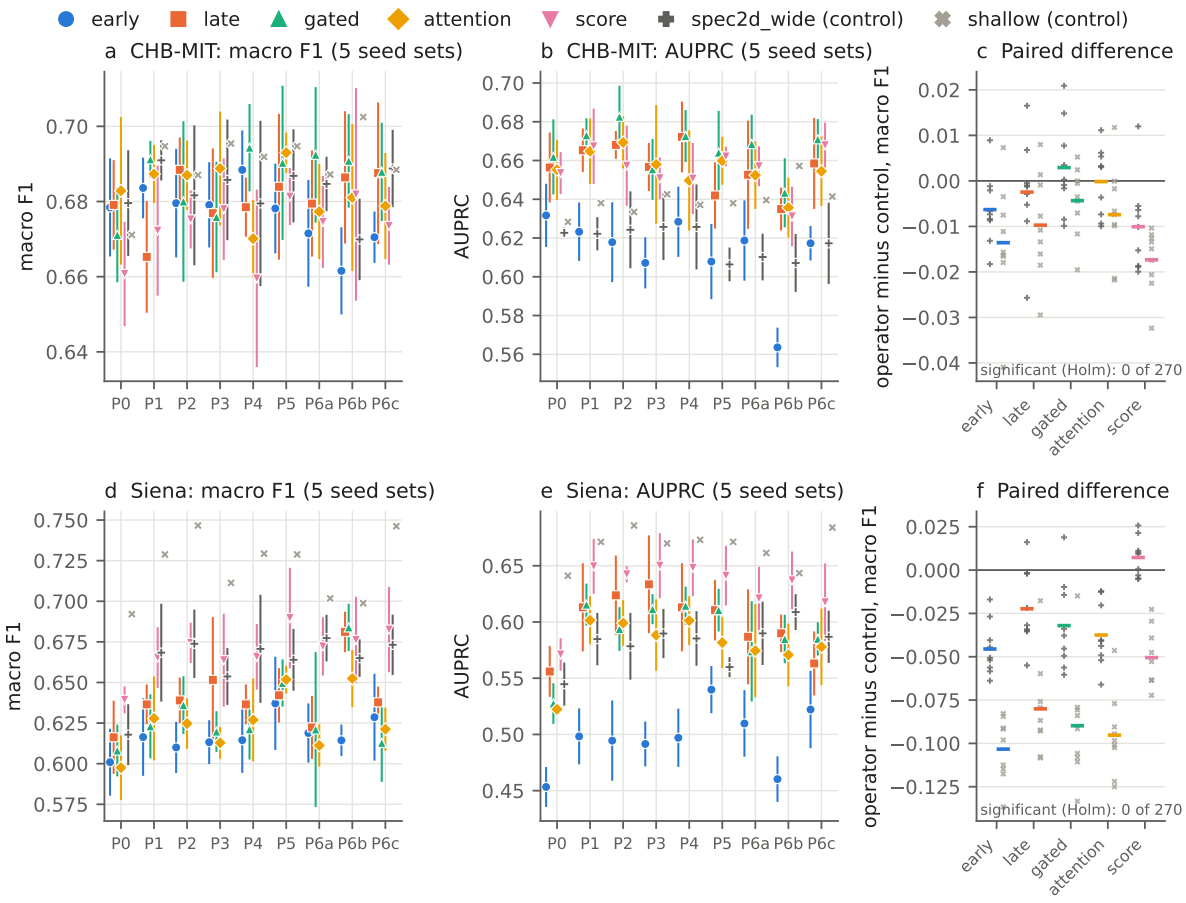


Figure 2: Fusion operators against the controls. **a, b, d, e**, Macro F1 and AUPRC of each operator and of two controls (the spectrogram network of the fusion budget, spec2d_wide, and the shallow logistic regression) in each preprocessing pipeline, on CHB-MIT (a, b; 23 LOSO folds) and Siena (d, e; 14 folds); mean and SD over five seed sets, each run averaged over its folds. **c, f**, Paired difference in macro F1 between each operator and each control, one point per pipeline (fold values averaged over seed sets), with the mean as a bar; none of the 270 control-operator comparisons per dataset was significant after Holm correction.

non-significant result as equivalence would have hidden this distinction. Supplementary Table S8 gives every paired difference with its confidence interval, corrected p and equivalence margin.

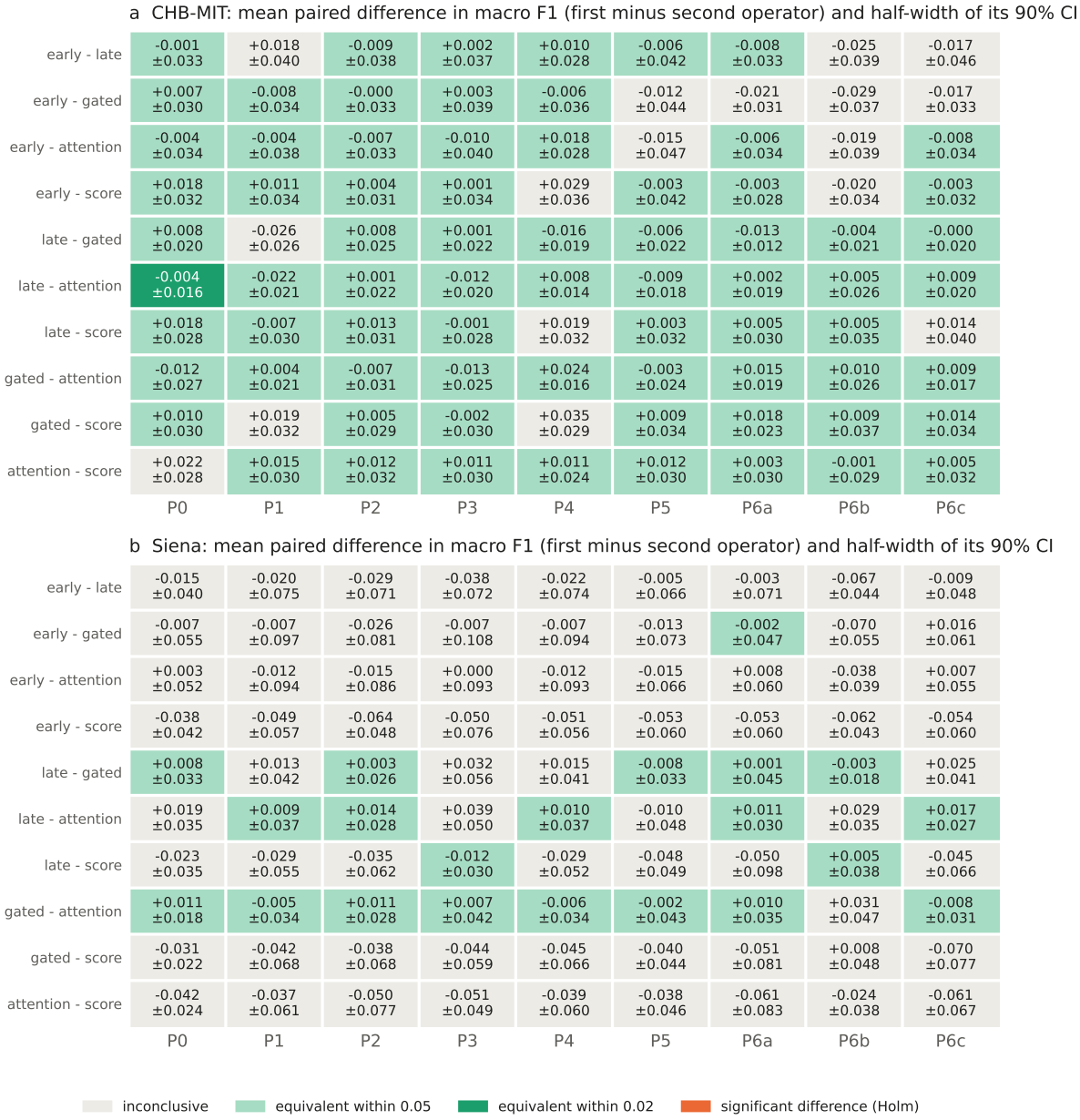


Figure 3: Equivalence matrix. For every operator pair (rows) in every preprocessing pipeline (columns): the mean paired difference in macro F1 (first minus second operator, fold values averaged over five seed sets) and the half-width of its 90 percent confidence interval with the corrected variance; the colour gives the outcome of the two one-sided tests at the margins of ±0.05 and ±0.02 and of the Holm-corrected test of a difference. **a**, CHB-MIT. **b**, Siena.

3.3 Operator rankings are sensitive to the seed and to preprocessing decisions

Although the mean differences were modest, which operator ranked first varied substantially between runs (Figure 4).

On CHB-MIT gated fusion ranked first in 38 percent of runs, followed by attention (24), late (20), early (11) and score fusion (7), and every operator ranked last in at least 13 percent of runs; no operator kept a consistently superior position. Changing only the seed reordered the

operators almost completely: Kendall's W across the five seed sets of one pipeline had a median of 0.22 (0.14 to 0.62 over pipelines), and rankings from different seeds of the same pipeline agreed with a mean Kendall tau of 0.10. Rankings from different pipelines agreed with a mean tau of 0.03; the difference did not exceed what permuting the pipeline labels produces ($p = 0.073$), and among the three artefact removal pipelines alone there was no indication that changing the algorithm reordered the operators more than changing the seed (difference -0.07, $p = 0.88$). The window-level predictions changed about as much between pipelines as between seeds (Cohen's kappa 0.56 and 0.58).

On Siena the rankings were more consistent. Score fusion ranked first in 82 percent of runs and never last (mean rank 1.29); Kendall's W had a median of 0.58 (0.40 to 0.78), and rankings agreed with a mean tau of 0.37 between seeds and 0.34 between pipelines ($p = 0.074$). This consistency did not establish significant superiority: score fusion's advantage over the next operator, 0.030 macro F1 on average, lies within the range that fourteen subjects can neither confirm nor exclude, and the operator main effect was $F = 2.87$ (uncorrected $p = 0.032$, Greenhouse-Geisser $p = 0.092$, partial eta squared 0.18). An exception emerged among the three artefact removal pipelines: on Siena, changing the cleaning algorithm reordered the operators more than changing the seed (difference in mean Kendall tau 0.18, $p = 0.029$). The relative importance of preprocessing and of stochastic training therefore depends on the dataset and on the preprocessing decision compared.

The sensitivity of statistical inference to the number of seeds is illustrated by the Fusion × Preprocessing interaction on CHB-MIT. With three seed sets the Greenhouse-Geisser-corrected test gave $p = 0.034$; with five it gave $p = 0.069$ ($F = 1.73$, uncorrected $p = 0.008$, epsilon 0.34; partial eta squared 0.073). This change should not be read as an interaction that disappeared, but as a demonstration that conclusions drawn from a few training runs can depend on which seeds were included. On Siena there was no interaction with either number of seed sets ($p =$ 0.36 and 0.39).

3.4 Subject heterogeneity dominates the variability due to model and preprocessing choices

To identify the factors behind the variability, each metric was decomposed into subject, operator, preprocessing, seed and interaction components (Figure 5a, b).

On CHB-MIT the held-out subject accounted for 74.2 percent of the variance of macro F1 (SD 0.126), its interactions with pipeline and operator for 7.2 percent and the residual for 17.3 percent. All seed terms together accounted for 1.1 percent (SD 0.015), more than the operator and preprocessing, whose estimates were below zero (-2.4e-5 and -2.7e-5), that is, below their error terms (a zero share means that these effects could not be separated from their error terms, not that they are absent); the Fusion × Preprocessing component was 0.09 percent, consistent with the small practical size of the interaction despite its sensitivity to the number of seeds. AUPRC (subject 76.8 percent, operator 0.8, seed 1.5) and balanced accuracy (72.9, 0, 1.4) showed the same structure. Preprocessing had no detectable main effect in the repeated-measures ANOVA either ($F = 0.29$, Greenhouse-Geisser $p = 0.83$, partial eta squared 0.013).

On Siena the variability was more distributed. The subject accounted for 32.5 percent and its interactions with pipeline and operator for a further 26.9 percent; the operator held 1.5 percent of the variance of macro F1 (SD 0.016) and 5.5 percent of AUPRC, the seed 4.2 percent and preprocessing 0.06 percent (main effect $F = 1.09$, $p = 0.36$).

The effects of fusion and preprocessing therefore cannot be characterised by their dataset-level averages alone. Mean macro F1 per held-out subject ranged from 0.46 (chb05) to 0.87 (chb22) on CHB-MIT and from 0.51 (PN07) to 0.78 (PN00) on Siena (Figure 5c, d). Within a subject the operators differed by a median of 0.063 macro F1 on CHB-MIT and 0.10 on Siena, three and

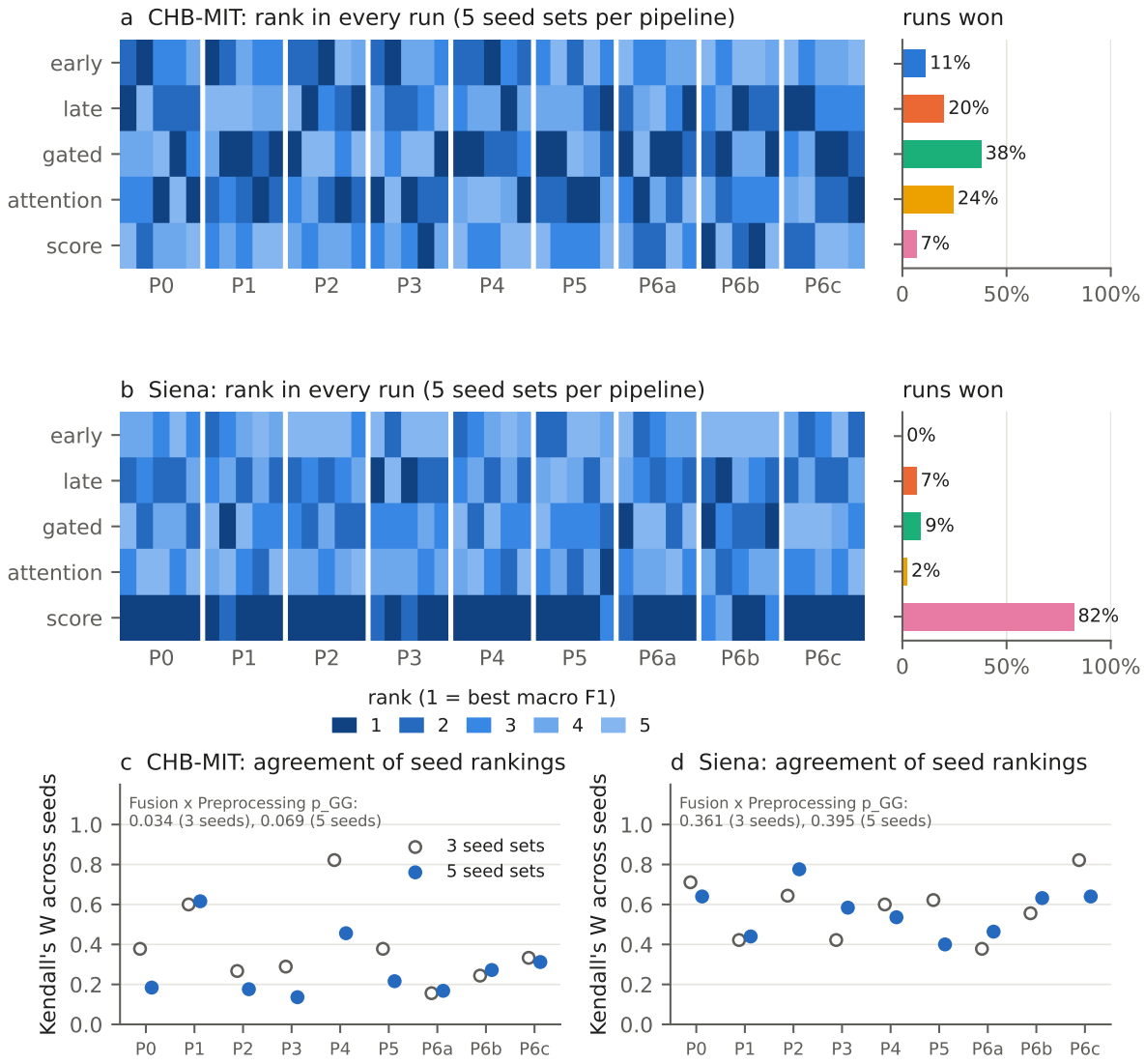


Figure 4: Ranking stability. **a, b**, Rank of each operator by macro F1 in every run (within each pipeline, one column per seed set; darker is better), and the share of the 45 runs of each dataset in which the operator ranked first. **c, d**, Agreement of the operator rankings across seed sets within each pipeline (Kendall's W) with three and with five seed sets, and the Greenhouse-Geisser-corrected p of the Fusion × Preprocessing interaction with each.

two times the median difference within a pipeline, and the best operator changed from subject to subject: on CHB-MIT each of the five operators was the best for at least two subjects. An operator that suits one subject often does not suit another. This heterogeneity also explains why small differences in mean macro F1 were difficult to establish statistically: with 23 subjects on CHB-MIT and 14 on Siena, the uncertainty of every comparison is dominated by the variation between held-out subjects. The variance decomposition thus identifies subject heterogeneity as a central feature of the benchmark rather than as residual noise.

3.5 Artefact removal involves trade-offs between signal preservation, performance, cost and reproducibility

The three artefact removal procedures (Extended Infomax, GEDAI and AMICA) were evaluated for their effect on classification and also for signal preservation, computational cost and reproducibility (Figure 6).

Infomax and AMICA removed a median of one component per recording and kept a median of 78 percent of the signal's power; GEDAI kept 99.9 percent in most recordings but removed more than half of the power in 21 percent of them. They differed strongly in cost: cleaning the 670 CHB-MIT recordings took 8.5 CPU hours with Infomax, 4.7 with GEDAI and 73 with AMICA. Infomax and AMICA removed mostly delta power and, within a recording, slightly less from ictal than from non-ictal windows (a median difference of 0.010 and 0.017 of the power kept, Wilcoxon $p < 0.001$). Over the whole corpus, GEDAI's default preset kept a median of 63 percent of the power of CHB-MIT's ictal windows against 93 percent of the non-ictal ones, because the recordings it cleaned heavily held many of the seizures; within recordings it showed no difference (Supplementary Table S5).

The effect on classification was not uniform across operators or datasets. Against P1, the same signal without artefact removal, late fusion gained 0.014 to 0.022 macro F1 on CHB-MIT under all three algorithms and early fusion lost 0.012 to 0.022; on Siena GEDAI raised late and gated fusion by 0.045 and 0.061. AUPRC fell under every algorithm for most operators, by up to 0.060 (early fusion under GEDAI, CHB-MIT). No algorithm was better than the others for every operator and setting.

A separate analysis showed that the procedures differed in reproducibility (Section 2.7). With the same code, package versions and seeds, GEDAI cleaned about 27 percent of recordings differently on a second machine. Extended Infomax removed different components in about 2.5 percent of recordings between machines and in about 1.2 percent between two runs on the same machine with different process parallelism. AMICA removed the same components in every recording on both machines, and its retained power differed by at most 0.003. Experiments with a third machine of another architecture and with rounding-level perturbations of the input suggest that some cleaning decisions are sensitive to floating-point variation near the algorithms' decision thresholds.

These differences were not confined to intermediate signals. Scoring the same trained models on recordings cleaned on two machines changed seizure-level sensitivity by up to 0.054 under GEDAI and changed the order of the operators by sensitivity (Kendall tau 0.60 to 1.00); the change was at most 0.005 under Infomax and absent under AMICA. Reproducibility at the level of code, package versions and seeds therefore did not guarantee reproducible cleaned signals or downstream predictions. The results identify a trade-off: an inexpensive cleaning method can introduce more variation between machines, whereas a costlier one can give more stable outputs, and the reproducibility of preprocessing can affect the conclusions of a model comparison.

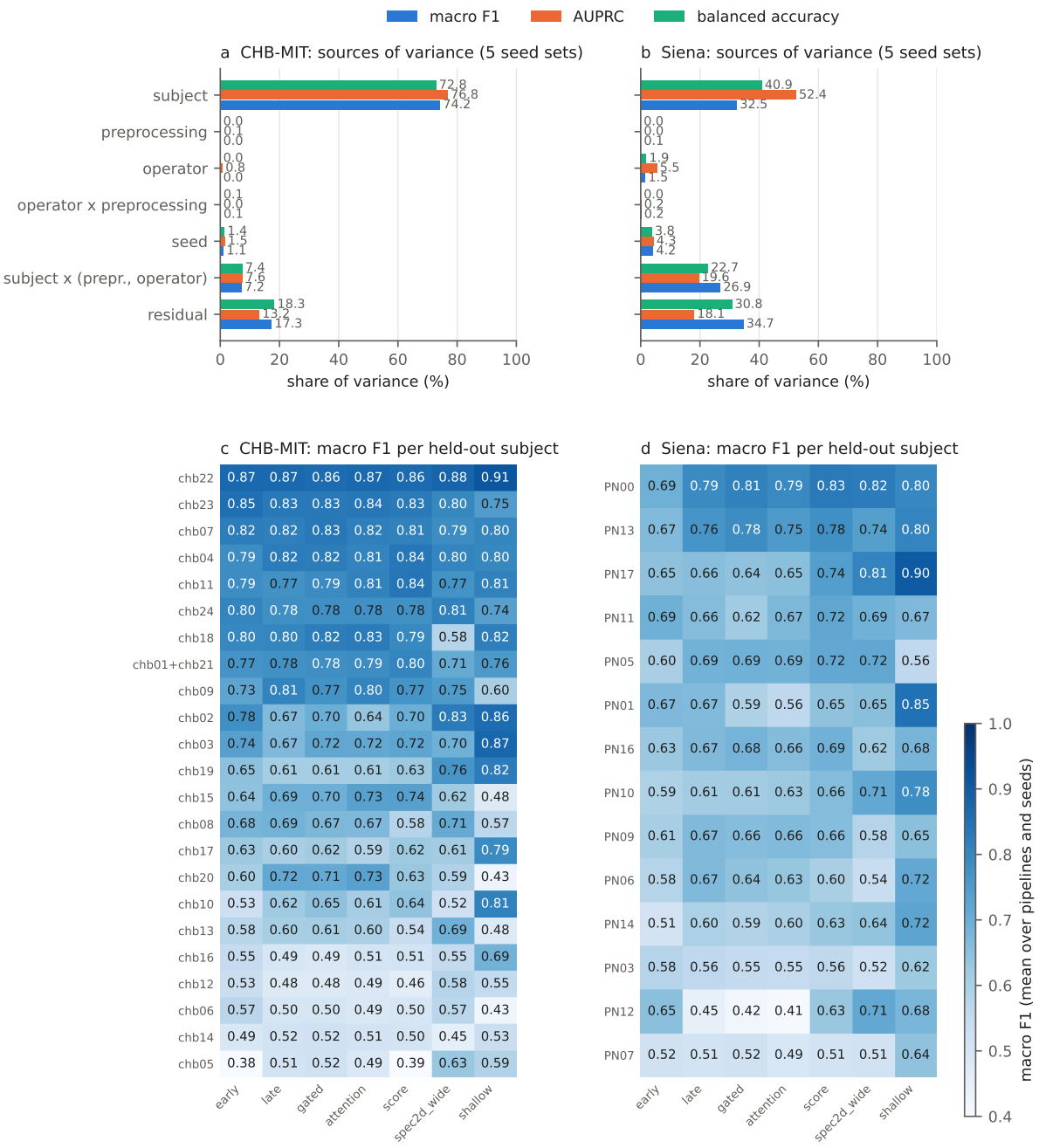


Figure 5: Sources of variability. **a, b**, Shares of the variance of macro F1, AUPRC and balanced accuracy (method-of-moments estimates for a fully crossed random-effects model of subject, preprocessing, operator and seed; negative estimates shown as 0; seed: all seed terms together). **c, d**, Macro F1 per held-out subject and model, mean over the nine pipelines and five seed sets; subjects sorted by their mean over the models.

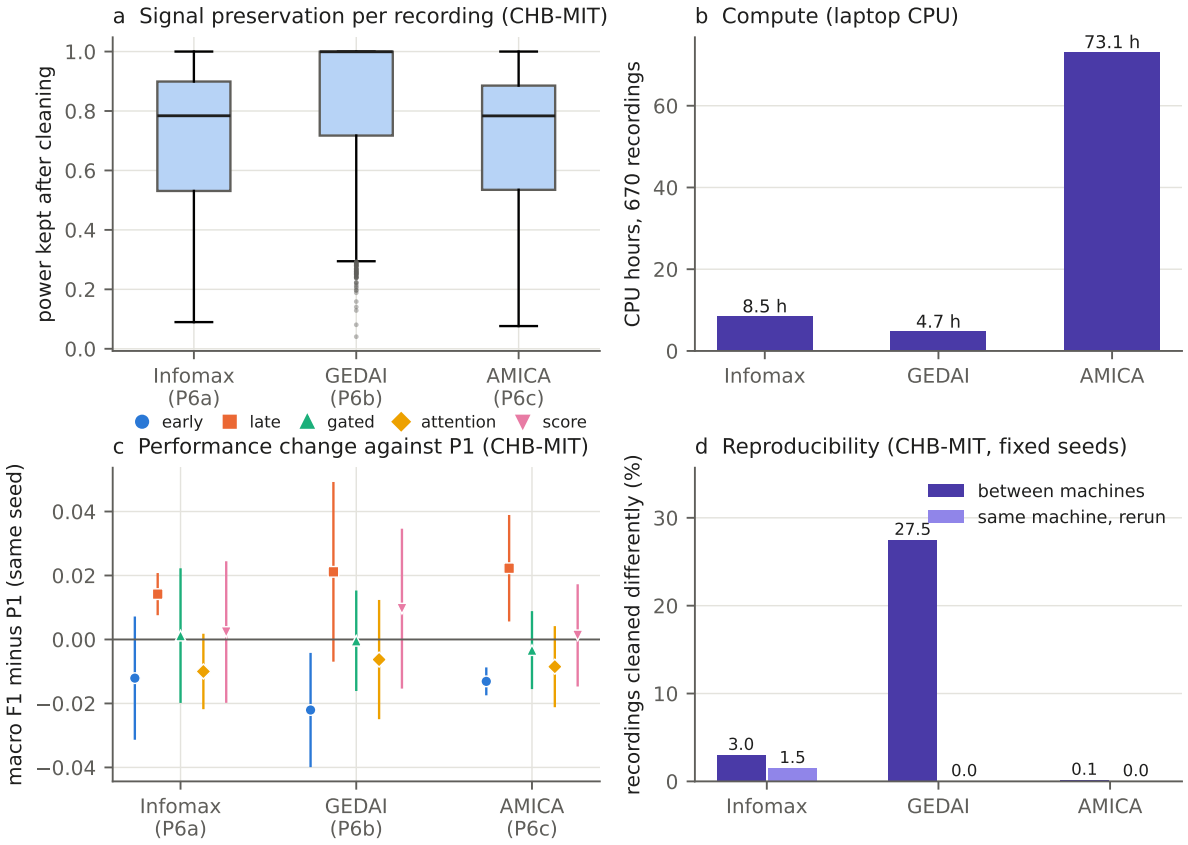


Figure 6: Artefact removal in P6 (CHB-MIT). **a**, Share of each recording's power kept after cleaning (670 recordings). **b**, CPU hours for the 670 recordings on a laptop CPU. **c**, Change in macro F1 against P1, the same 0.5 to 40 Hz signal without artefact removal, paired by seed set; mean and SD over five seed sets. **d**, Share of recordings cleaned differently (different components removed, or retained power differing by more than 0.001) on a second machine and in a second run on the same machine, with the same code, package versions and seeds.

3.6 Window-level differences translate into different seizure-level operating points

The window-level analysis was complemented by seizure-level evaluation on the complete recordings, with the probability threshold of 0.5 fixed in advance and without post-processing (Figure 7).

On CHB-MIT the operators detected 68 to 79 percent of the annotated seizures, with 5.2 to 7.7 false alarms per hour. Early fusion had a distinct operating point: it reached the highest mean seizure-level sensitivity (0.79) and ranked first in sensitivity in 82 percent of runs, but it also raised the most false alarms in every run (7.7 per hour against 5.2 to 5.7 for the other operators). Taken together with the window-level results, in which early fusion had the lowest AUPRC in every pipeline of both datasets, this shows that a higher sensitivity at a given threshold does not imply better discrimination: early fusion labels more windows ictal, which detects more seizures at the price of more false alarms.

Artefact removal did not remove this trade-off. Under Infomax and AMICA the operators detected 66 to 76 percent of seizures with 4.8 to 7.2 false alarms per hour; under GEDAI, 71 to 80 percent with 5.5 to 8.7. On Siena the operators detected 84 to 87 percent of seizures, without a consistent order; false alarms per hour are not reported for Siena because its recordings are cut around the seizures and do not represent continuous monitoring.

Small differences in window-level macro F1 can therefore coexist with clearly different balances between seizure sensitivity and false-alarm burden, and window-level classification metrics alone do not characterise how a seizure detector would operate.

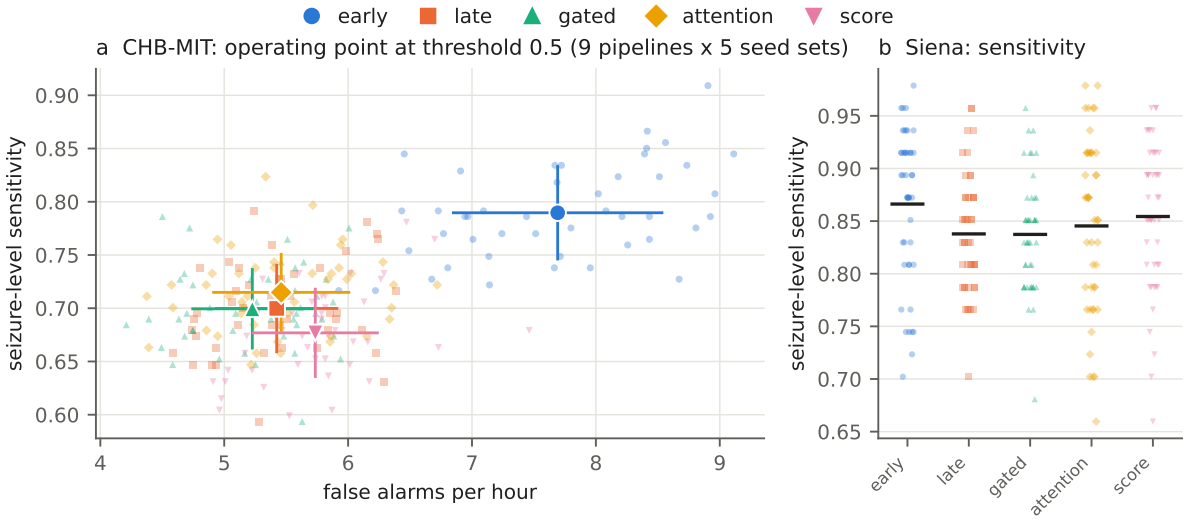


Figure 7: Seizure-level detection at the fixed threshold of 0.5 without post-processing. **a**, CHB-MIT: sensitivity against false alarms per hour, one point per run (nine pipelines × five seed sets) and the operator's mean and SD. **b**, Siena: seizure-level sensitivity per operator, one point per run, with the mean as a bar; false alarms per hour are not reported for Siena.

3.7 Predictive performance transfers across datasets, but operator rankings do not

Finally, we examined whether the patterns observed in subject-independent evaluation generalise across datasets (Figure 8).

Models trained on all of CHB-MIT and tested on Siena reached a mean macro F1 of 0.641 (attention) to 0.677 (late) over pipelines and seed sets. For early, late, gated and attention fusion this was higher than the mean obtained by training and testing on Siena (LOSO); score fusion was the exception, with 0.652 after transfer against 0.670 within Siena.

Comparable aggregate performance did not mean that the order of the operators carried over. Score fusion, first in 82 percent of Siena's own runs, was first in 11 percent of the transfer runs; there, late fusion ranked first most often (38 percent) and attention fusion was most often last (40 percent). Kendall tau between the seed-averaged operator rankings of the same pipeline was -0.11 on average between CHB-MIT and Siena, 0.07 between CHB-MIT and the transfer, and -0.11 between Siena and the transfer.

These results distinguish two forms of generalisation: whether a trained model keeps useful performance on an independent dataset, and whether the advantage of one fusion operator over another is preserved. The former was observed at the level of aggregate macro F1, the latter was not; an operator that ranked highly when trained and evaluated on one dataset did not necessarily keep that position under external evaluation.

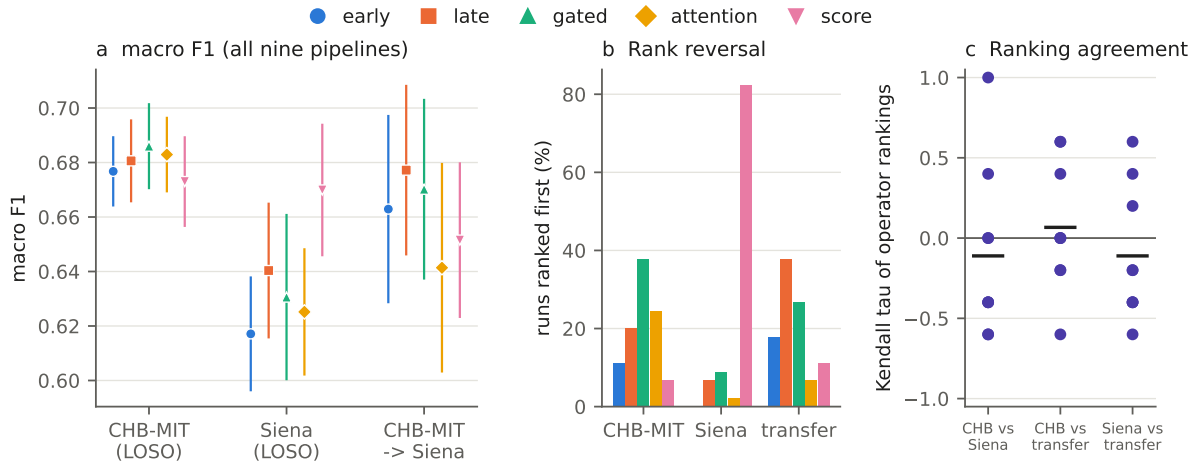


Figure 8: Cross-dataset generalisation. **a**, Macro F1 of each operator on CHB-MIT (LOSO), on Siena (LOSO), and trained on CHB-MIT and tested on Siena; mean and SD over the runs of all nine pipelines and five seed sets. **b**, Share of runs in which each operator ranks first in each setting. **c**, Kendall tau between the seed-averaged operator rankings of the same pipeline in two settings; one point per pipeline (points can overlap), bar: mean.

3.8 Sensitivity of fusion conclusions to decision thresholds and class prevalence

The primary analyses used a fixed decision threshold of 0.5 and a sampled non-ictal-to-ictal window ratio of 4:1. We therefore examined whether the conclusions about fusion operators were preserved when these evaluation choices were varied. These analyses were conducted post hoc and are interpreted as exploratory sensitivity assessments (Section 2.6; Figure 9; Supplementary Tables S9 and S10).

At the window level, changing the decision threshold from 0.5 to 0.3 or 0.7 generally preserved the absence of statistically significant differences between fusion operators, although their relative rankings changed across individual runs. Of the 540 comparisons (two datasets, nine pipelines, three thresholds, ten pairs), two were significant under the threshold-specific correction, both in the CHB-MIT P4 pipeline at a threshold of 0.7: gated fusion exceeded attention fusion by 0.021 macro F1 (raw $p = 0.004$, Holm-adjusted $p = 0.034$) and score fusion by 0.066 (raw $p = 0.002$, adjusted $p = 0.019$). Under the additional across-threshold correction neither remained

significant (adjusted p = 0.111 and 0.058; Supplementary Table S9c). The operator with the highest macro F1 in a run differed from the one at 0.5 in 27 percent of the CHB-MIT runs at a threshold of 0.3 and in 33 percent at 0.7, and in 53 and 29 percent of the Siena runs; on Siena, score fusion ranked first in 82 percent of the runs at 0.5, 38 percent at 0.3 and 67 percent at 0.7 (Figure 9a; Supplementary Table S9b). Thus, the broad conclusion that no fusion operator consistently outperformed the others was comparatively stable, whereas the identity of the highest-ranked operator in an individual run was sensitive to the operating threshold.

The threshold dependence also affected comparisons with the classical baseline. On CHB-MIT, the shallow classifier had the lowest mean macro F1 among the evaluated models at a threshold of 0.3 (0.642), but the highest at 0.7 (0.715). On Siena, it remained the highest-performing model by mean macro F1 at all three thresholds (0.687, 0.720 and 0.725). These findings indicate that conclusions about the value of architectural complexity depend partly on the decision rule used to convert model scores into class predictions.

At the seizure level, the additional evaluation of P0 and P1 on CHB-MIT showed that the operator with the highest seizure sensitivity in a run was relatively stable across thresholds on CHB-MIT (the same as at 0.5 in 7 of 10 runs at 0.3 and in 9 of 10 at 0.7, but on Siena in only 5 and 7 of 10), whereas the false-alarm ranking was more sensitive: the operator with the fewest false alarms was the same as at 0.5 in all 10 runs at 0.3 but in only 1 of 10 at 0.7 (Figure 9b; Supplementary Table S10a). Score fusion was particularly threshold-dependent, with its sensitivity falling from 0.81 to 0.51 and its false-alarm rate from 8.3 to 3.3 per hour between thresholds of 0.3 and 0.7, whereas early fusion had the highest sensitivity of the operators at every threshold. These results show that stability in seizure sensitivity does not necessarily imply stability in false-alarm burden. Because the additional event-level analyses were restricted to P0 and P1, their conclusions should not be generalised to all nine preprocessing pipelines.

Reweighting the existing CHB-MIT predictions to the non-ictal window counts of the complete recordings, a prevalence of 0.36 percent ictal windows instead of 20 percent, reduced the mean macro F1 of the fusion operators from 0.673 to 0.686 under the sampled evaluation to 0.517 to 0.562 under prevalence adjustment (subject-level aggregation; Figure 9c; Supplementary Table S10b). The decline follows from the change in prevalence itself: the weighting leaves sensitivity unchanged, but each false positive now stands for 15 to 334 non-ictal windows, so precision at the 0.5 threshold, and with it the ictal-class F1, fell for every model. The shallow classifier, which had the highest sampled macro F1 (0.690), showed the largest decline in both absolute and relative terms (to 0.454, a decline of 0.237 or 34 percent, against 0.124 to 0.160 or 18 to 24 percent for the operators) and had the lowest prevalence-adjusted macro F1; pooled aggregation gave the same order (operators 0.483 to 0.500, shallow 0.447). At 0.5 it labelled 21.5 percent of the non-ictal test windows as ictal, against 7.4 to 11.0 percent for the operators, which under the adjusted prevalence left it a precision of 0.02 (operators 0.11 to 0.24) at a sensitivity of 0.69. Its prevalence-adjusted AUPRC was nevertheless the highest of all models (0.295 against 0.221 to 0.283). Because AUPRC summarises precision over all thresholds, this suggests that the shallow classifier did not rank windows worse and that another threshold might have reduced its loss; it does not show that the fixed threshold alone caused the decline, since the lower prevalence reduces precision at every threshold, and no threshold was chosen under the adjusted prevalence.

Operator pairs were compared on the prevalence-adjusted macro F1 with the procedure of the primary analysis: the corrected resampled t test on the 23 per-subject fold values (seed sets averaged; 22 degrees of freedom) and Holm adjustment within the ten operator pairs of each pipeline. Of the 90 comparisons, 22 had a raw p below 0.05 and three remained significant after Holm adjustment, all with early fusion lower: in P3 against late fusion (difference −0.042; corrected 95 percent interval −0.069 to −0.014; raw p = 0.004, Holm-adjusted p = 0.040) and against attention fusion (−0.057; −0.091 to −0.023; p = 0.002, adjusted p = 0.021), and in P5 against gated fusion (−0.068; −0.106 to −0.029; p = 0.001, adjusted p = 0.014; Supplementary

Table S10d). No pair differed after adjustment in prevalence-adjusted AUPRC (5 of 90 with a raw p below 0.05). The Holm adjustment controls the error rate within each pipeline, not across the nine pipelines, and the analysis was specified after the primary results were known; the three results are therefore exploratory findings rather than evidence that early fusion is inferior, although they agree with its weaker ranking of windows (Section 3.1). The operator ranked first under prevalence adjustment was the one ranked first under the sampled evaluation in 42 percent of the runs. This analysis demonstrates that absolute performance estimates, and potentially the apparent advantages of particular models, are sensitive to the evaluation class distribution. However, these estimates rely on the representativeness of the sampled non-ictal windows and should not be interpreted as directly measured performance on the complete continuous recordings.

The subject-level bootstrap, which does not model the overlap of the LOSO training sets, gave narrower intervals than the corrected test. Compared at the same nominal level, without adjustment for multiplicity, the bootstrap 95 percent interval excluded zero for 10 of the 90 operator pairs on CHB-MIT and 24 on Siena (3 and 17 when the seed sets were also resampled), and the corrected 95 percent interval for 1 and 6 (Supplementary Table S10c); after the Holm adjustment of the primary analysis, no pair was significant. The nominal bootstrap intervals are therefore not evidence of significant differences. The two procedures also rest on different assumptions: the corrected test approximates the dependence created by the shared training data with the Nadeau-Bengio variance inflation and assumes approximately normal per-subject differences, whereas the percentile bootstrap treats the held-out subjects as independent draws and, with 23 and 14 subjects, may give intervals that are too narrow. On Siena, pooled over pipelines, the nominal bootstrap intervals of the differences between score fusion and each jointly trained operator excluded zero, whereas the nominal corrected interval did so only for early fusion; the direction of these differences was the same in at least 98 percent of the resamples (Supplementary Table S10c). The direction of score fusion's advantage on Siena is therefore consistent, while whether it is statistically supported depends on the dependence assumptions of the inference.

Taken together, the post hoc analyses reinforce the distinction between model performance, model ranking, and the robustness of a superiority claim. The primary inference was broadly preserved under alternative evaluation choices, but the preferred operator and the magnitude of its apparent advantage could depend on the threshold, the metric, and the assumed class prevalence.

4. Discussion

This study examined whether conclusions about multimodal EEG fusion operators remain defensible when the analytical conditions surrounding their comparison are systematically varied. Rather than optimising a single seizure detector, we evaluated the robustness of comparative conclusions across preprocessing pipelines, stochastic training, subject-independent evaluation, datasets and computational environments. The principal finding is that architectural differences alone did not establish a consistently superior fusion strategy. Although some operators achieved higher mean performance in particular settings, these advantages were generally small, statistically uncertain, or unstable across evaluation conditions. The additional post hoc analyses further showed that decision thresholds and class-prevalence assumptions could change model rankings and absolute performance estimates. Together, these findings suggest that the validity of a fusion superiority claim depends as much on the evaluation design as on the architecture being compared.

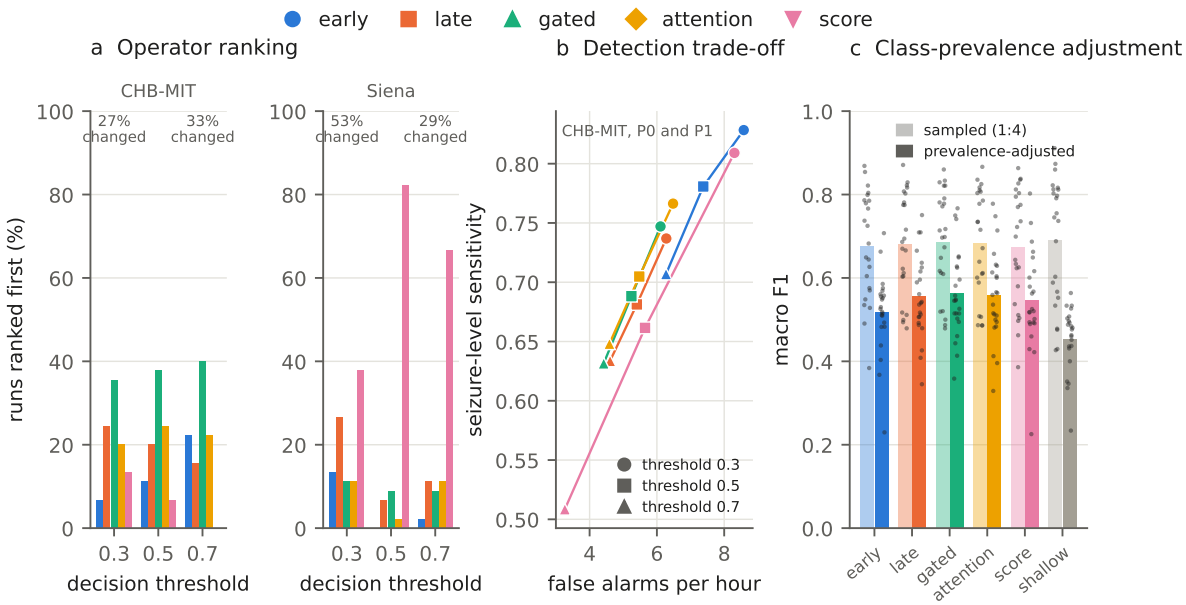


Figure 9: Sensitivity of model rankings and performance to post hoc evaluation choices. **a**, Frequency with which each fusion operator ranked first (highest fold-mean macro F1 of a run; ties share the first place) at decision thresholds of 0.3, 0.5 and 0.7 in CHB-MIT and Siena, over the 45 runs of each dataset (nine pipelines, five seed sets); the annotation gives the share of runs whose best operator differs from that at 0.5. **b**, Seizure-level sensitivity versus false alarms per hour for the five fusion operators across the same thresholds in CHB-MIT pipelines P0 and P1; mean over the ten runs (two pipelines, five seed sets), event counts pooled over the held-out persons of a run; lines connect the thresholds of one operator. **c**, Macro F1 under the original 1:4 sampled evaluation and after prevalence adjustment using the non-ictal window counts of the full CHB-MIT recordings, for the fusion operators and the shallow classifier; bars, mean over the 23 held-out persons; dots, each person's macro F1 averaged over the 45 runs. All analyses are exploratory and were performed after completion of the primary experiments. Prevalence-adjusted results are estimates based on reweighting existing predictions, not direct evaluations of all non-ictal windows.

4.1 Fusion complexity does not establish robust superiority

The comparison of parameter-matched jointly trained fusion operators, together with score fusion, did not reveal a consistent performance advantage for a particular fusion mechanism. Across preprocessing pipelines, the median difference between the highest- and lowest-performing operators was 0.021 macro F1 on CHB-MIT and 0.053 on Siena, and no pairwise difference survived the corrected comparisons; the operator's share of the variance was estimated below zero on CHB-MIT and at 1.5 percent on Siena. Although score fusion frequently ranked first on Siena, its advantage was not statistically established under the corrected comparisons. This distinction is important because a consistently favourable mean or ranking does not necessarily demonstrate that an operator provides a reliable performance improvement.

The single-branch and classical controls provide additional context. A spectrogram-only network with the parameter budget of the fusion models achieved a mean macro F1 within the range of the jointly trained operators on CHB-MIT and above all of them on Siena, while a logistic regression on seven hand-crafted features per channel remained competitive on both datasets at the prespecified threshold, though not in AUPRC on CHB-MIT. None of these differences was significant, and few were demonstrably equivalent: the controls stood to the operators as the operators stood to each other. These findings do not establish that simple models are universally preferable, nor do they rule out benefits from fusion under different training regimes. Instead, they indicate that increasing architectural complexity should be justified against appropriately controlled alternatives rather than assumed to improve seizure detection, in line with a broader evaluation of multimodal models, available as a preprint, in which added fusion complexity often did not pay off (Rheude et al. 2025).

One comparatively consistent architectural pattern emerged: early fusion produced the lowest AUPRC in all nine preprocessing pipelines of both datasets. Yet its window-level macro F1 remained comparable to that of the other operators, and it achieved the highest seizure-level sensitivity of the operators at the cost of additional false alarms. Early fusion is the only operator that combines the two representations before either branch has learned its own features, and its weaker ranking of windows is the one property of an operator that the multiverse preserved. This divergence illustrates why classification performance at a single threshold, probability ranking and event-level detection should not be treated as interchangeable indicators of model quality.

The results therefore support a more restrained interpretation of fusion architecture comparisons. The relevant question is not simply which operator achieves the highest mean score, but whether its advantage persists under matched capacity, appropriate controls, alternative evaluation conditions and statistical uncertainty.

4.2 Statistical significance, practical equivalence and ranking stability answer different questions

A central methodological contribution of this study is the explicit separation of statistical significance, practical equivalence and ranking stability. These properties answer related but distinct questions. Statistical significance concerns whether the observed difference is sufficiently supported under the assumed sampling and dependence structure. Equivalence testing asks whether differences are small enough to fall within a prespecified, practically acceptable margin. Ranking stability examines whether the relative order of operators persists across analytical choices.

The three approaches did not always produce the same interpretation. On CHB-MIT, 73 of the 90 operator pairs (ten pairs in nine pipelines) satisfied the broader equivalence margin of 0.05 macro F1, but only one the stricter margin of 0.02. On Siena, 21 of 90 comparisons supported equivalence within 0.05, and none within 0.02, despite the absence of statistically significant

differences. The data therefore rule out large differences between operators on CHB-MIT, but not differences of the size that published comparisons typically report as improvements, and a nonsignificant comparison cannot be interpreted as proof that two operators perform equivalently.

Ranking analyses provided another perspective. On CHB-MIT, agreement across seed sets was low, with a median Kendall's W of 0.22, indicating that the relative order of operators was sensitive to stochastic training; at the differences observed here, a ranking obtained from one training run says little about the ranking under another seed. Rankings from different pipelines agreed about as poorly as rankings from different seeds of one pipeline (mean Kendall tau 0.03 against 0.10; permutation $p = 0.073$), and the Fusion $\times$ Preprocessing interaction, significant with three seed sets, was no longer significant with five (Greenhouse-Geisser $p = 0.069$), with a variance share of 0.09 percent. Preprocessing therefore mostly added variation rather than a systematic advantage for one operator; the one exception was the choice among the three artefact removal algorithms on Siena, which reordered the operators more than the seed did ($p = 0.029$). Conversely, score fusion ranked first in 82 percent of the Siena runs without establishing a significant advantage. A stable ranking can therefore coexist with uncertain performance differences, while practically similar operators may exchange ranks across runs; when two operators are equally good, their order is decided by noise.

The dependence between leave-one-subject-out folds further complicates inference. Treating these folds as independent observations would understate uncertainty because the training sets overlap extensively: in a simulation of two equivalent classifiers the naive paired test rejected in 37.6 percent of cases at a nominal 5 percent. The resampling-corrected paired comparisons were therefore retained as the primary inferential approach. The post hoc subject-level bootstrap, which treats subjects as independent, gave narrower intervals, and its nominal intervals, not adjusted for multiplicity, excluded zero for some pairs, notably score fusion's advantage on Siena, whose direction was consistent across resamples (Section 3.8). These nominal intervals are comparable with the nominal corrected intervals, which excluded zero for fewer pairs, not with the Holm-adjusted primary results, under which no pair was significant. The two procedures rest on different dependence assumptions, which reinforces the importance of reporting inferential sensitivity rather than treating one statistical procedure as definitive.

These findings suggest that comparative EEG studies should report the magnitude and uncertainty of performance differences, prespecified equivalence margins where appropriate, and rank distributions across repeated runs. None of these quantities can substitute for the others: a stable ranking is not evidence of a meaningful difference, a significant effect is not evidence of a large one, and the absence of a significant difference is not evidence of equivalence.

4.3 Subject heterogeneity and dataset shift limit generalisability

Variation associated with the held-out subject was a major contributor to observed performance differences, accounting for approximately 74 percent of the estimated variance on CHB-MIT and, together with its interactions with pipeline and operator, about 60 percent on Siena. This should not be interpreted as a purely physiological effect. The subject-associated component also reflects differences in seizure characteristics, recording equipment and conditions, seizure counts, and other properties that vary across held-out individuals. Nevertheless, its magnitude indicates that subject-level heterogeneity was substantially more influential than the estimated main effects of fusion operator or preprocessing pipeline in this experimental design.

This heterogeneity has direct implications for model comparison. The interaction of subject with pipeline and operator, 7 percent of the variance on CHB-MIT and 27 percent on Siena, means that an operator that performs favourably on one collection of subjects may not maintain the same advantage on another, even when the preprocessing and training protocol are held fixed. Averaging performance across subjects can therefore conceal important differences in the

conditions under which a particular model succeeds or fails, and with 23 and 14 subjects the variation between them also bounds the precision of every comparison: the equivalence bounds of Section 3.2 are as wide as they are because the subjects differ so much.

Cross-dataset evaluation provided a complementary test of generalisability. Models trained on CHB-MIT achieved aggregate macro F1 on Siena comparable to that of models trained on Siena itself, but the ranking of fusion operators was not preserved (mean Kendall tau between -0.11 and 0.07). Score fusion, for example, ranked first in 82 percent of the within-Siena runs but in only 11 percent of the CHB-MIT-to-Siena transfer runs. Part of its advantage on Siena is inherited from its spectrogram branch: the default-width spectrogram network alone (0.660 macro F1; 0.663 at the fusion budget) already had a higher mean on Siena than every jointly trained operator, and score fusion averages it with the waveform network using a weight chosen on Siena's validation subjects. Whether end-to-end training of the joint operators overfits Siena's 14 subjects, which training the branches independently would avoid, was not tested. Thus, the transferability of aggregate predictive performance should not be confused with the transferability of conclusions about the best architecture.

Because CHB-MIT and Siena differ simultaneously in patient populations, acquisition characteristics, seizure distributions and recording protocols, the observed changes cannot be attributed to any single source of domain shift. The findings instead demonstrate the limitations of treating architectural rankings obtained from one dataset as generalisable properties of the models themselves.

4.4 Evaluation choices shape model selection and clinical interpretation

The post hoc sensitivity analyses extended the main findings by examining whether model rankings and performance estimates were preserved under alternative decision thresholds and class-prevalence assumptions. Although the overall absence of significant operator differences was largely maintained, and the two comparisons that reached significance at one threshold did not survive the correction across thresholds, the identity of the highest-ranked operator varied across individual runs. This indicates that even when the broad statistical conclusion is comparatively stable, the practical decision about which model to select may depend on the chosen operating threshold.

The classical baseline provided a particularly informative example. On CHB-MIT, its relative position changed from last at a threshold of 0.3 to first at 0.7, whereas it remained the highest-performing model by mean macro F1 at all tested thresholds on Siena. This finding does not imply that one threshold is intrinsically preferable. Rather, it demonstrates that comparisons between classical and deep-learning approaches are conditional on the decision rule applied to their outputs.

Event-level evaluation further distinguished seizure detection from window classification. In the primary analysis, early fusion achieved the highest seizure sensitivity of the operators (0.79) but also generated the most false alarms (7.7 per hour); which of these is preferable is a clinical choice between missed seizures and alarm fatigue (Sendelbach and Funk 2013), not a property that window-level metrics reveal. The additional threshold analysis of P0 and P1 showed that sensitivity rankings could remain comparatively stable while false-alarm rankings changed. Such trade-offs are central to the practical interpretation of seizure detection systems, where detecting additional seizures may come at the cost of an increased alarm burden.

The class-prevalence analysis identified a separate limitation of balanced or artificially sampled evaluation sets. Reweighting the existing CHB-MIT predictions to approximate the full non-ictal window distribution reduced the estimated macro F1 of the fusion operators from about 0.67 to 0.69 to about 0.52 to 0.56, and that of the shallow classifier, the highest under the sampled

evaluation, to 0.45, the lowest. At a fixed threshold, the much lower prevalence reduces precision for every model, and most for those that label more non-ictal windows as ictal, as the shallow classifier did at 0.5. Its prevalence-adjusted AUPRC, the highest of all models, indicates that it did not rank windows worse, but does not show that another threshold would have removed the decline. Three operator pairs differed after Holm adjustment, all with early fusion lower; as a post hoc result controlled only within each pipeline, this is exploratory. Because this calculation assumes that the sampled non-ictal windows represent the excluded windows, it should be interpreted as a prevalence-adjusted sensitivity estimate rather than a direct assessment of continuous-monitoring performance. Nevertheless, it demonstrates that reported performance, and the apparent advantage of one model over another, can be strongly influenced by the evaluation class distribution.

These findings also clarify the clinical scope of the present study. The models were evaluated without patient-specific threshold optimisation or post-processing, were trained only on other people's seizures, and used offline rather than causal preprocessing. At the prespecified threshold the operators detected 68 to 79 percent of CHB-MIT's seizures with 5.2 to 7.7 false alarms per hour, that is 125 to 185 false alarms per day of monitoring, and no person reached a mean below one false alarm per hour. Generic scalp EEG detectors evaluated for clinical use have reported 0.1 to 5 false alarms per hour, most about 0.2 to 0.3, and patient-specific detectors fewer; a low false alarm rate is regarded as essential for clinical acceptance, and validation standards require it to be reported (Baumgartner et al. 2018; Beniczky and Ryvlin 2018). The person mattered more than the choice of operator: mean sensitivity ranged from 0.24 (chb06) to 1.00 across persons and the false alarm rate from 1.3 to 12.3 per hour, while the operators' means spanned 0.68 to 0.79 and 5.2 to 7.7, which motivates patient-specific adaptation, not tested here, rather than the search for a better generic operator; patient-specific classifiers detected 96 percent of 173 test seizures in CHB-MIT with a median of two false detections per day (Shoeb and Guttag 2010). Siena cannot answer the question of false alarms, because its recordings are cut around seizures, and neither dataset contains the long, artefact-rich ambulatory or intensive-care recordings in which detectors are used. The observed false-alarm rates therefore do not support claims of clinical readiness. The purpose of the study is not to recommend a deployable seizure detector, but to demonstrate why model selection based on a single threshold or window-level metric may be insufficient.

4.5 Computational reproducibility is part of methodological validity

A further contribution of the study concerns the reproducibility of EEG artefact removal. Even with fixed random seeds and software versions, preprocessing outcomes were not fully preserved across computational environments. GEDAI produced different cleaning outcomes for approximately 27 percent of CHB-MIT recordings when repeated on another machine, while Extended Infomax changed its component decision in 2.5 percent of recordings between machines and in 1.2 percent between two runs on one machine. AMICA removed the same components in every recording under the evaluated conditions.

These differences were not merely numerical discrepancies without downstream consequences. Applying the same trained models to differently cleaned signals changed seizure-level sensitivity by up to 0.054 under GEDAI and, in some runs, the relative ranking of fusion operators. Perturbing the input by one part in ten million reproduced similar decision changes for recordings close to cleaning thresholds, which suggests sensitivity to numerical perturbations near threshold decisions (a component's $|z|$ against an EOG proxy, GEDAI's noise threshold) as a plausible explanation. However, the experiments do not isolate a unique numerical mechanism, and no causal attribution to a particular machine, library or hardware component is warranted. Neuroimaging pipelines, including ICA decompositions of fMRI data, are known to give different results on different

operating systems (Glatard et al. 2015); we are not aware of a comparable report for EEG artefact removal, where algorithms can be deterministic within a machine (GEDAI), or identical within one process but not between two runs with different process parallelism (Infomax), and still not reproducible across machines.

The CPU–GPU comparison similarly showed that identical seeds and nominally identical training configurations did not guarantee interchangeable performance estimates. The observed mean absolute macro F1 difference between CPU and GPU execution was 0.074, slightly larger than the 0.066 difference between the compared CPU seed sets. The prespecified comparison criterion was therefore narrowly missed. This result should be reported as an empirical limitation of cross-device reproducibility rather than as evidence that one execution environment is more accurate; for this reason all training runs of one analysis used the same device.

Together, these observations indicate that computational reproducibility requires more than recording software versions and random seeds. For preprocessing methods involving decomposition, component selection or threshold-dependent decisions, retaining cleaning logs, component decisions and processed outputs may be essential for reproducing downstream conclusions, and the repository of this study holds them per recording. Importantly, the present window-level results were obtained from a consistent set of cleaned signals; the cross-environment experiments characterise a reproducibility limitation rather than invalidate those primary comparisons.

4.6 Implications, limitations and recommendations for future EEG fusion studies

The findings suggest that EEG fusion studies should evaluate the robustness of comparative conclusions rather than report only the highest performance achieved under one configuration. A defensible comparison should include appropriately matched architectural controls, simple baseline models, subject-independent evaluation, repeated stochastic runs, and explicit treatment of dependence between evaluation folds. Where feasible, it should also examine preprocessing sensitivity, practical equivalence, ranking stability, threshold-dependent operating points and computational reproducibility. Table 5 summarises these elements as practical reporting recommendations drawn from the present results, not as a new standard.

Table 5: Recommended robustness reporting elements for EEG fusion comparisons.

Evaluation dimension	Minimum reporting requirement	Rationale	Evidence from the present study
Subject-independent evaluation	Hold out whole persons, all recordings of one individual together; select validation subjects from the training subjects only	Windows of one person are correlated, so mixing them across partitions inflates performance	Subject-associated variability made up 74 percent of the variance on CHB-MIT; two CHB-MIT cases are one person (Sections 2.1, 3.4)
Parameter and baseline comparability	Report parameter counts; include a size-matched single-branch network and a simple classical baseline	A gain may reflect capacity or the representation rather than fusion	A spectrogram network of the same size and a 127-parameter logistic regression matched or exceeded the operators at the prespecified threshold (Section 3.1)
Multiple training seeds	Train each configuration with several seed sets and report the spread	A single run confounds the operator with the seed	Median Kendall's W across five seed sets 0.22 on CHB-MIT (Section 3.3)
Preprocessing sensitivity	Repeat the comparison under several defensible pipelines	A winner under one pipeline may not hold under another	Rankings agreed between pipelines about as poorly as between seeds; artefact removal reordered the operators on Siena (Section 3.3)

Evaluation dimension	Minimum reporting requirement	Rationale	Evidence from the present study
Corrected statistical inference	Account for overlapping training sets in cross-validation and for multiple comparisons	Naive paired tests overstate significance	The naive test rejected in 37.6 percent of simulated comparisons of equivalent classifiers at a nominal 5 percent (Section 2.6)
Practical equivalence	Prespecify an equivalence margin and report the smallest margin the data support	A nonsignificant difference is not evidence of equivalence	73 and 21 of 90 pairs equivalent within 0.05 on CHB-MIT and Siena, 1 and 0 within 0.02 (Section 3.2)
Ranking stability	Report rank distributions across runs rather than one ranking	Similar operators exchange ranks between runs	Score fusion ranked first in 82 percent of Siena runs without a significant advantage, and in 11 percent after transfer (Sections 3.3, 3.7)
Threshold and event-level sensitivity	Report more than one operating point, and seizure-level sensitivity and false alarms per hour	Rankings and the standing of a baseline depend on the decision rule	The best operator changed in 27 to 53 percent of runs between thresholds; the false-alarm ranking changed at 0.7 (Section 3.8)
Class-prevalence assumptions	State the class ratio of the evaluation and report prevalence-adjusted or full-recording estimates	Sampled ratios inflate precision-dependent metrics	Prevalence-adjusted macro F1 of about 0.52 to 0.56 against 0.67 to 0.69 sampled (Section 3.8)
Computational reproducibility	Report machines and devices; store cleaning logs, component decisions or cleaned signals	Threshold decisions in artefact removal may not reproduce across machines	GEDAI cleaned 27 percent of recordings differently on a second machine, changing seizure-level sensitivity by up to 0.054 (Section 3.5)

Several limitations qualify the generality of the present findings. First, the study included 23 CHB-MIT subjects and 14 Siena subjects, limiting the precision of subject-level comparisons and equivalence conclusions; many pairs were neither different nor demonstrably equivalent, and five seed sets bound the seed variance only roughly. Second, the evaluated models shared one backbone family, one pair of EEG representations (raw waveform and log-magnitude STFT) and a restricted parameter budget; different architectures, representations or optimisation strategies may exhibit other patterns. Third, a fixed training configuration was used without architecture-specific hyperparameter optimisation. This improves comparability but does not establish the best achievable performance of each operator, and an operator might gain more than another from tuning.

Fourth, the primary window-level evaluation used an artificially sampled 1:4 non-ictal-to-ictal ratio, and a window was labelled ictal if it overlapped a seizure at all. The prevalence-adjusted analysis provides an informative sensitivity check but does not replace direct evaluation of all continuous-recording windows. Fifth, the additional threshold and bootstrap analyses were post hoc and should not be interpreted as independent confirmatory evidence. Sixth, event-level threshold sensitivity was evaluated only for P0 and P1, while Siena's seizure-centred recordings do not support a meaningful estimate of false alarms per hour in continuous monitoring. Seventh, preprocessing was offline: filters and artefact cleaning see the complete recording, so the results do not describe a causal, real-time detector. The artefact removal settings (16 components, frontopolar channels as EOG proxies, no muscle rejection) and the projection of GEDAI's reference onto a bipolar montage are our choices for montages without EOG, and GEDAI was designed for referential, high-density recordings. Finally, the cross-dataset experiment combines multiple sources of domain shift and cannot identify their individual contributions. Further details concern the data: 13 of CHB-MIT's 198 seizures lie in recordings without the standard montage and were not used; Siena's summed recording time (141.0 h by the EDF headers) differs

from the dataset description (128.4 h) for reasons that remain unexplained for three patients; and the title and abstract screening of the systematic literature search was done by an AI assistant (agreement with an independent AI pass, kappa 0.81), with the human screening and the full texts of the remaining candidates in progress, so that claims marked [abstract] in Section 1 rest on abstracts.

5. Conclusions

This study asked whether a fusion operator that appears superior in one EEG seizure-detection experiment remains superior when the analytical conditions around the comparison change. For RQ1, the differences between the parameter-matched operators were small, with median within-pipeline spreads of 0.021 and 0.053 macro F1 on CHB-MIT and Siena, and no pair differed significantly under the corrected comparisons in any pipeline; on CHB-MIT most pairs were equivalent within 0.05 but almost none within 0.02, and on Siena most were neither different nor demonstrably equivalent. For RQ2, the operator ranked first changed with the seed, the preprocessing pipeline, the decision threshold, the assumed class prevalence and the machine on which artefact removal ran, and a stable first place, as for score fusion on Siena, did not come with a significant advantage. For RQ3, aggregate performance transferred from CHB-MIT to Siena, but the ranking of the operators did not. Throughout, the variability associated with the held-out subject outweighed the effects of the operator and of preprocessing.

Within these boundaries, the study provides evidence that selecting a fusion operator on the basis of one dataset, preprocessing pipeline, training run or operating threshold can produce conclusions that are not preserved under other defensible analytical choices. A model may achieve the highest score without providing sufficient evidence of robust superiority. Superiority should therefore be demonstrated through the stability, uncertainty and practical relevance of comparative results, not inferred from a single favourable performance estimate.

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Supplementary Material

Table S1: Artefact removal in P6: signal retention and compute. Per recording of CHB-MIT with selected windows (670), on the 0.5 to 40 Hz signal. Power kept = signal power after removal / before. Extended Infomax and AMICA share 16 components, fitting on every 8th sample and the same automatic ocular component rule; GEDAI selects what to remove by its own criterion. Compute on a laptop CPU (Intel Core Ultra 7 155U), six recordings in parallel.

method	recordings	power kept, median	IQR	min	share < 0.5	components removed, mean	seconds per recording, median	CPU hours, total
P6a Extended Infomax	670	0.784	0.531-0.899	0.090	0.222	1.491	33	8.5
P6b GEDAI, conservative (used)	670	0.999	0.717-0.999	0.041	0.206		18	4.7
P6c AMICA	670	0.784	0.535-0.885	0.076	0.228	1.531	213	73.1
GEDAI, default (not used)	670	0.658	0.317-0.999	0.041	0.433		22	5.6

Table S2: GEDAI preset, chosen on signal preservation only. Six CHB-MIT subjects (chb01, chb03, chb05, chb08, chb12, chb15), the same windows under both presets. Power kept per recording and per window (relative to the same 0.5 to 40 Hz signal); no classification result was used for the choice (`src/gedai_qc.py`).

preset	unit	n	p5	p25	median	p75	share < 0.5	share < 0.1
default (auto)	recording	199	0.197	0.324	0.843	0.999	0.387	0.000
default (auto)	ictal window	591	0.241	0.367	0.779	1.000	0.333	0.005
default (auto)	non-ictal window	2364	0.263	0.395	0.999	0.999	0.328	0.003
conservative (auto-), used	recording	199	0.263	0.957	0.999	0.999	0.166	0.000
conservative (auto-), used	ictal window	591	0.313	0.999	1.000	1.000	0.134	0.000
conservative (auto-), used	non-ictal window	2364	0.331	0.999	0.999	1.000	0.126	0.002

Table S3: P4: technical artefact rejection per person. A window is rejected if a channel is flat (standard deviation below 1 μV) or holds one constant value for at least 0.5 s, on the unfiltered signal. Rejected windows are removed from training and validation only; test windows are unchanged. Rejection rates are given for ictal and non-ictal windows separately. chb01 includes chb21, the same person.

person	windows	ictal	ictal rejected	non-ictal	non-ictal rejected	ictal rejected %	non-ictal rejected %
chb01	370	74	0	296	1	0.0	0.3
chb02	100	20	2	80	2	10.0	2.5
chb03	225	45	7	180	1	15.6	0.6
chb04	210	42	5	168	5	11.9	3.0
chb05	310	62	0	248	0	0.0	0.0
chb06	120	24	1	96	0	4.2	0.0
chb07	180	36	2	144	1	5.6	0.7
chb08	485	97	4	388	3	4.1	0.8
chb09	155	31	3	124	0	9.7	0.0
chb10	245	49	11	196	4	22.4	2.0
chb11	415	83	1	332	0	1.2	0.0
chb12	615	123	1	492	7	0.8	1.4

person	windows	ictal	ictal rejected	non-ictal	non-ictal rejected	ictal rejected %	non-ictal rejected %
chb13	335	67	0	268	5	0.0	1.9
chb14	120	24	0	96	0	0.0	0.0
chb15	1070	214	0	856	1	0.0	0.1
chb16	70	14	0	56	1	0.0	1.8
chb17	160	32	0	128	6	0.0	4.7
chb18	190	38	1	152	0	2.6	0.0
chb19	135	27	3	108	0	11.1	0.0
chb20	180	36	1	144	2	2.8	1.4
chb22	120	24	0	96	0	0.0	0.0
chb23	245	49	0	196	0	0.0	0.0
chb24	325	65	0	260	0	0.0	0.0
all	6380	1276	42	5104	39	3.3	0.8

Table S4: Siena: metadata problems and the decisions taken. Every departure from the published seizure lists, with the decision used in the main analysis, its kind (a correction verified against the recording, or a decision rule applied without such a check), how it was checked, and how much the alternative would change the windows. Other, purely textual fixes (a space inside a time in PN10's list, misspelt file names, PN12's missing registration times taken from the same file's first block, recordings past midnight read modulo 24 h) are listed in `docs/SIENA.md`, Section 4. Recording time is taken from the EDF headers, 141.0 h; the dataset's description gives 128.4 h, a difference explained for PN14 (the start time above) but not for PN03, PN10 and PN12, whose files contain no padding (`docs/SIENA.md`, Section 8).

item	recording / seizure	problem in the metadata	used	type
1	PN00-3, seizure 3	listed end 19.29.29 lies 1916 s after the recording ends (18.57.33)	end read as 18.29.29 (a 60 s seizure)	verified correction
2	PN05-3, seizure 3	registration start: list 06.01.23, EDF header 06.01.13	EDF header start time, as for every recording	decision rule (not a verified correction)
3	PN10-3, seizure 3	two onsets: 15.43.53 (clinical), 15.43.59 (electrical)	electrical onset (advisor)	decision
4	PN10-4.5.6, seizure 6	only a clinical onset (15.18.26)	used as listed, flagged in the corpus's seizure notes	decision
5	PN10-2, seizure 2	end "11.41.04 opure 11.40.43" (Italian "or")	first value, 11.41.04	decision rule
6	PN14-3	registration start: list 16.17.45, EDF header 19.17.45	EDF header start time	verified correction

How each decision was checked, and its effect:

item	how it was checked	remaining uncertainty	effect on the windows
1	signal: the right frontotemporal 3-20 Hz ictal rhythm (FP2-F8, F8-T8) falls below a quarter of its peak 14 s after 18.29.29; the patient's other four seizures end 0 to 13 s after their listed ends by the same rule	offset from an automatic rule on one channel pair	alternative (exclude the seizure and the rest of the recording): 30 instead of 37 ictal windows for PN00

item	how it was checked	remaining uncertainty	effect on the windows
2	none possible: the header defines the first sample and 39 of 41 recordings agree with their list; the list's value fits the 20 s end offset of 18 other recordings, the header's does not	10 s in the seizure's position	4 ictal windows either way, shifted by one: 2 window labels differ
3	CHB-MIT annotates electrographic onsets; the electrical onset keeps the two corpora consistent	none (both listed)	clinical onset instead: 48 instead of 47 ictal windows for PN10 (1 label)
4	-	electrical onset unknown; in seizure 3 it followed the clinical one by 6 s	at most about one window, by analogy with seizure 3
5	-	21 s in the seizure's end	second value instead: 45 instead of 47 ictal windows for PN10
6	header start + duration (41,995 s) gives the listed end time 06.57.40 exactly; the list's start would make the recording three hours longer than the file	none	seizure at 1 h 52 min into the recording, not 4 h 52 min

Table S5: Artefact removal: power kept per window, by band and class. What each method removes from the 0.5 to 40 Hz signal, per 10 s window (cleaned power divided by the power of the same window before cleaning, summed over channels), for ictal and non-ictal windows, overall and in five bands (delta 0.5-4, theta 4-8, alpha 8-13, beta 13-30, low gamma 30-40 Hz; medians). The second part compares the classes within recordings: per recording the median of each class, their difference, and a Wilcoxon signed-rank test over the recordings that hold both. A difference between the classes over the whole corpus that is absent within recordings reflects which recordings a method cleans heavily, not a preference for one class (`src/p6_signal_effects.py`). GEDAI's default preset was applied to CHB-MIT only.

dataset	method	windows	n	power kept, median	share < 0.5	delta	theta	alpha	beta	low gamma
chbmit	Extended Infomax	ictal	1276	0.872	0.092	0.855	0.905	0.929	0.933	0.935
chbmit	Extended Infomax	non-ictal	5104	0.854	0.187	0.835	0.893	0.911	0.927	0.933
chbmit	GEDAI conservative (used)	ictal	1276	1.000	0.134	0.999	1.000	1.000	1.000	1.000
chbmit	GEDAI conservative (used)	non-ictal	5104	0.999	0.167	0.999	1.000	1.000	1.000	1.000
chbmit	GEDAI default (not used)	ictal	1276	0.625	0.448	0.583	0.739	0.836	0.882	0.909
chbmit	GEDAI default (not used)	non-ictal	5104	0.925	0.372	0.920	0.966	0.983	0.991	0.994
chbmit	AMICA	ictal	1276	0.873	0.093	0.851	0.909	0.922	0.929	0.933
chbmit	AMICA	non-ictal	5104	0.839	0.191	0.821	0.882	0.901	0.921	0.930
siena	Extended Infomax	ictal	328	0.682	0.271	0.609	0.759	0.847	0.900	0.941
siena	Extended Infomax	non-ictal	1312	0.657	0.370	0.520	0.714	0.862	0.906	0.914
siena	GEDAI conservative (used)	ictal	328	0.679	0.427	0.694	0.745	0.850	0.833	0.944

dataset	method	windows	n	power kept, median	share < 0.5	delta	theta	alpha	beta	low gamma
siena	GEDAI conservative (used)	non-ictal	1312	0.997	0.383	0.997	1.000	1.000	1.000	1.000
siena	AMICA	ictal	328	0.723	0.232	0.660	0.823	0.887	0.929	0.959
siena	AMICA	non-ictal	1312	0.719	0.334	0.627	0.800	0.915	0.936	0.939

Within recordings:

dataset	method	recordings	ictal - non-ictal, median	share with ictal lower	Wilcoxon p
chbmit	Extended Infomax	137	0.0098	0.2847	0.0001
chbmit	GEDAI conservative (used)	137	0.0001	0.3212	0.0016
chbmit	GEDAI default (not used)	137	-0.0000	0.5036	0.0363
chbmit	AMICA	137	0.0170	0.2847	0.0000
siena	Extended Infomax	41	-0.0050	0.5122	0.6769
siena	GEDAI conservative (used)	41	-0.0000	0.5610	0.6815
siena	AMICA	41	0.0000	0.4390	0.8790

Table S6: Window-level performance per pipeline and operator. Mean and SD over the five seed sets of each run's fold mean, for every pipeline and fusion operator (`python -m src.make_results`, `results_v2/results/main_*.csv`). **CHB-MIT, macro F1** (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.678 ±0.013	0.679 ±0.012	0.671 ±0.013	**0.683** ±0.020	0.661 ±0.014
P1	0.684 ±0.008	0.665 ±0.015	**0.691** ±0.005	0.687 ±0.008	0.672 ±0.017
P2	0.680 ±0.014	**0.688** ±0.009	0.680 ±0.021	0.687 ±0.009	0.675 ±0.008
P3	0.679 ±0.011	0.677 ±0.017	0.676 ±0.015	**0.689** ±0.015	0.678 ±0.014
P4	0.688 ±0.010	0.678 ±0.008	**0.694** ±0.012	0.670 ±0.010	0.660 ±0.024
P5	0.678 ±0.012	0.684 ±0.019	0.690 ±0.021	**0.693** ±0.005	0.681 ±0.008
P6a	0.672 ±0.014	0.679 ±0.014	**0.692** ±0.018	0.677 ±0.013	0.675 ±0.012
P6b	0.662 ±0.012	0.686 ±0.018	**0.691** ±0.012	0.681 ±0.020	0.682 ±0.028
P6c	0.671 ±0.007	0.688 ±0.019	**0.688** ±0.013	0.679 ±0.014	0.674 ±0.010

CHB-MIT, AUPRC (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.632 ±0.016	0.656 ±0.018	**0.662** ±0.019	0.655 ±0.015	0.654 ±0.011
P1	0.623 ±0.015	0.665 ±0.011	**0.673** ±0.009	0.665 ±0.017	0.667 ±0.019
P2	0.618 ±0.021	0.668 ±0.007	**0.683** ±0.016	0.669 ±0.013	0.657 ±0.021
P3	0.607 ±0.013	0.657 ±0.012	0.655 ±0.016	**0.658** ±0.031	0.651 ±0.011
P4	0.628 ±0.018	0.672 ±0.018	**0.673** ±0.013	0.650 ±0.026	0.651 ±0.018
P5	0.608 ±0.019	0.642 ±0.017	**0.664** ±0.022	0.660 ±0.012	0.662 ±0.005
P6a	0.619 ±0.021	0.653 ±0.028	**0.668** ±0.015	0.652 ±0.017	0.657 ±0.010
P6b	0.564 ±0.010	0.635 ±0.011	**0.643** ±0.018	0.636 ±0.015	0.631 ±0.015
P6c	0.617 ±0.009	0.658 ±0.023	**0.671** ±0.010	0.654 ±0.018	0.668 ±0.012

CHB-MIT, balanced accuracy (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	**0.684** ±0.009	0.681 ±0.010	0.674 ±0.010	0.684 ±0.016	0.667 ±0.014
P1	0.684 ±0.006	0.669 ±0.010	**0.685** ±0.008	0.684 ±0.008	0.680 ±0.015
P2	0.684 ±0.010	0.682 ±0.006	0.679 ±0.022	**0.685** ±0.009	0.678 ±0.010
P3	**0.687** ±0.010	0.676 ±0.015	0.676 ±0.012	0.687 ±0.013	0.683 ±0.011
P4	**0.690** ±0.008	0.677 ±0.006	0.687 ±0.008	0.667 ±0.011	0.663 ±0.020
P5	0.686 ±0.015	0.683 ±0.014	0.684 ±0.019	**0.694** ±0.008	0.687 ±0.008
P6a	0.676 ±0.014	0.675 ±0.014	**0.688** ±0.017	0.680 ±0.006	0.679 ±0.014
P6b	0.670 ±0.009	0.682 ±0.017	**0.688** ±0.012	0.680 ±0.016	0.686 ±0.017
P6c	0.677 ±0.008	0.682 ±0.017	**0.686** ±0.009	0.680 ±0.015	0.681 ±0.010

Siena, macro F1 (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.601 ±0.021	0.616 ±0.023	0.608 ±0.016	0.598 ±0.020	**0.639** ±0.008
P1	0.616 ±0.024	0.637 ±0.012	0.623 ±0.020	0.628 ±0.026	**0.665** ±0.019
P2	0.610 ±0.016	0.639 ±0.014	0.636 ±0.018	0.625 ±0.016	**0.674** ±0.012
P3	0.613 ±0.014	0.652 ±0.039	0.620 ±0.013	0.613 ±0.010	**0.664** ±0.029
P4	0.614 ±0.020	0.637 ±0.012	0.621 ±0.019	0.627 ±0.025	**0.666** ±0.020
P5	0.637 ±0.029	0.642 ±0.017	0.650 ±0.015	0.652 ±0.008	**0.690** ±0.031
P6a	0.619 ±0.018	0.622 ±0.019	0.621 ±0.048	0.611 ±0.013	**0.672** ±0.018
P6b	0.614 ±0.010	0.681 ±0.012	**0.684** ±0.015	0.652 ±0.018	0.676 ±0.027
P6c	0.629 ±0.027	0.638 ±0.010	0.613 ±0.024	0.621 ±0.013	**0.683** ±0.026

Siena, AUPRC (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.453 ±0.018	0.556 ±0.023	0.527 ±0.018	0.522 ±0.003	**0.571** ±0.014
P1	0.498 ±0.025	0.613 ±0.039	0.615 ±0.019	0.601 ±0.021	**0.649** ±0.024
P2	0.494 ±0.036	0.624 ±0.035	0.594 ±0.019	0.599 ±0.020	**0.642** ±0.007
P3	0.492 ±0.020	0.633 ±0.043	0.611 ±0.013	0.588 ±0.032	**0.650** ±0.029
P4	0.497 ±0.026	0.613 ±0.039	0.614 ±0.017	0.601 ±0.022	**0.648** ±0.025
P5	0.540 ±0.021	0.611 ±0.027	0.610 ±0.019	0.582 ±0.023	**0.641** ±0.027
P6a	0.510 ±0.030	0.587 ±0.042	0.574 ±0.045	0.575 ±0.041	**0.621** ±0.028
P6b	0.460 ±0.020	0.590 ±0.017	0.585 ±0.022	0.571 ±0.028	**0.637** ±0.026
P6c	0.522 ±0.034	0.563 ±0.028	0.585 ±0.015	0.578 ±0.034	**0.618** ±0.034

Siena, balanced accuracy (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.621 ±0.021	0.663 ±0.018	0.652 ±0.011	0.639 ±0.013	**0.665** ±0.007
P1	0.643 ±0.022	0.676 ±0.014	0.673 ±0.018	0.673 ±0.024	**0.689** ±0.022
P2	0.645 ±0.024	0.680 ±0.008	0.684 ±0.012	0.674 ±0.014	**0.696** ±0.010
P3	0.641 ±0.018	**0.693** ±0.031	0.671 ±0.010	0.660 ±0.009	0.684 ±0.021
P4	0.643 ±0.022	0.676 ±0.014	0.673 ±0.018	0.673 ±0.025	**0.691** ±0.024
P5	0.661 ±0.023	0.688 ±0.015	0.691 ±0.011	0.687 ±0.010	**0.710** ±0.038
P6a	0.645 ±0.016	0.668 ±0.016	0.664 ±0.038	0.658 ±0.015	**0.682** ±0.011
P6b	0.622 ±0.011	0.688 ±0.017	**0.694** ±0.018	0.676 ±0.015	0.691 ±0.022
P6c	0.654 ±0.019	0.676 ±0.006	0.660 ±0.019	0.663 ±0.020	**0.697** ±0.018

Table S7: All controls. The six controls over the same complete runs as Table 4 (mean and SD over the 45 runs of each dataset, fold means within a run; Brier: lower is better). Parameters with 18 input channels; the classical models are logistic regressions on 18 log variances (logvar) and on seven features per channel (shallow), with an intercept (`results_v2/results/controls_*.csv`).

control	parameters	CHB-MIT macro F1	CHB-MIT AUPRC	CHB-MIT bal. acc.	CHB-MIT Brier	Siena macro F1	Siena AUPRC	Siena bal. acc.	Siena Brier
raw1d	28,850	0.663 ±0.014	0.604 ±0.019	0.678 ±0.013	0.332 ±0.018	0.609 ±0.034	0.565 ±0.040	0.659 ±0.024	0.443 ±0.061
spec2d	34,546	0.679 ±0.010	0.598 ±0.017	0.676 ±0.010	0.309 ±0.011	0.660 ±0.022	0.576 ±0.031	0.680 ±0.019	0.350 ±0.030
raw1d_wide	103,712	0.662 ±0.012	0.605 ±0.017	0.673 ±0.010	0.328 ±0.016	0.612 ±0.028	0.577 ±0.033	0.657 ±0.019	0.440 ±0.056
spec2d_wide	102,236	0.683 ±0.014	0.618 ±0.016	0.680 ±0.013	0.302 ±0.014	0.663 ±0.026	0.581 ±0.027	0.681 ±0.021	0.352 ±0.038
logvar	19	0.631 ±0.009	0.577 ±0.016	0.708 ±0.006	0.402 ±0.009	0.630 ±0.034	0.572 ±0.036	0.705 ±0.027	0.455 ±0.036
shallow	127	0.690 ±0.008	0.640 ±0.008	0.739 ±0.007	0.339 ±0.014	0.720 ±0.019	0.667 ±0.015	0.770 ±0.017	0.338 ±0.020

Table S8: Paired differences between operators, with confidence intervals and equivalence. For every operator pair in every pipeline: the mean paired difference in macro F1, its 90 percent confidence interval with the corrected variance (the interval used by the two one-sided tests), the corrected and the Holm-adjusted p of a difference, the smallest equivalence margin the data can establish ($\delta_{\min}$), and the outcome at the margins of ±0.05 and ±0.02 (`results_v2/multiverse/*_cuda/equivalence*.csv`; Figure 3). **CHB-MIT** (difference: first minus second operator, macro F1; fold values averaged over the seed sets)

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P0	early-late	-0.001	[-0.034, +0.033]	0.973	1.000	0.034	yes	no
P0	early-gated	+0.007	[-0.023, +0.037]	0.678	1.000	0.037	yes	no
P0	early-attention	-0.004	[-0.039, +0.030]	0.827	1.000	0.039	yes	no
P0	early-score	+0.018	[-0.014, +0.050]	0.353	1.000	0.050	yes	no
P0	late-gated	+0.008	[-0.012, +0.028]	0.491	1.000	0.028	yes	no
P0	late-attention	-0.004	[-0.020, +0.012]	0.692	1.000	0.020	yes	yes
P0	late-score	+0.018	[-0.009, +0.046]	0.266	1.000	0.046	yes	no
P0	gated-attention	-0.012	[-0.038, +0.015]	0.458	1.000	0.038	yes	no
P0	gated-score	+0.010	[-0.020, +0.041]	0.561	1.000	0.041	yes	no
P0	attention-score	+0.022	[-0.006, +0.050]	0.194	1.000	0.050	no	no
P1	early-late	+0.018	[-0.022, +0.059]	0.442	1.000	0.059	no	no
P1	early-gated	-0.008	[-0.041, +0.026]	0.701	1.000	0.041	yes	no
P1	early-attention	-0.004	[-0.041, +0.034]	0.866	1.000	0.041	yes	no
P1	early-score	+0.011	[-0.023, +0.046]	0.577	1.000	0.046	yes	no
P1	late-gated	-0.026	[-0.052, -0.000]	0.097	0.896	0.052	no	no
P1	late-attention	-0.022	[-0.043, -0.001]	0.090	0.896	0.043	yes	no
P1	late-score	-0.007	[-0.037, +0.023]	0.692	1.000	0.037	yes	no
P1	gated-attention	+0.004	[-0.017, +0.025]	0.752	1.000	0.025	yes	no
P1	gated-score	+0.019	[-0.013, +0.051]	0.326	1.000	0.051	no	no
P1	attention-score	+0.015	[-0.015, +0.045]	0.398	1.000	0.045	yes	no
P2	early-late	-0.009	[-0.046, +0.029]	0.689	1.000	0.046	yes	no
P2	early-gated	-0.000	[-0.034, +0.033]	0.982	1.000	0.034	yes	no
P2	early-attention	-0.007	[-0.041, +0.026]	0.705	1.000	0.041	yes	no
P2	early-score	+0.004	[-0.027, +0.035]	0.813	1.000	0.035	yes	no
P2	late-gated	+0.008	[-0.017, +0.034]	0.572	1.000	0.034	yes	no
P2	late-attention	+0.001	[-0.020, +0.023]	0.911	1.000	0.023	yes	no
P2	late-score	+0.013	[-0.018, +0.045]	0.478	1.000	0.045	yes	no
P2	gated-attention	-0.007	[-0.038, +0.024]	0.702	1.000	0.038	yes	no
P2	gated-score	+0.005	[-0.024, +0.034]	0.779	1.000	0.034	yes	no
P2	attention-score	+0.012	[-0.020, +0.044]	0.532	1.000	0.044	yes	no
P3	early-late	+0.002	[-0.035, +0.039]	0.919	1.000	0.039	yes	no
P3	early-gated	+0.003	[-0.036, +0.042]	0.889	1.000	0.042	yes	no
P3	early-attention	-0.010	[-0.050, +0.031]	0.684	1.000	0.050	yes	no
P3	early-score	+0.001	[-0.032, +0.035]	0.954	1.000	0.035	yes	no
P3	late-gated	+0.001	[-0.021, +0.023]	0.938	1.000	0.023	yes	no
P3	late-attention	-0.012	[-0.032, +0.008]	0.326	1.000	0.032	yes	no
P3	late-score	-0.001	[-0.029, +0.027]	0.948	1.000	0.029	yes	no
P3	gated-attention	-0.013	[-0.038, +0.012]	0.387	1.000	0.038	yes	no
P3	gated-score	-0.002	[-0.033, +0.028]	0.907	1.000	0.033	yes	no
P3	attention-score	+0.011	[-0.019, +0.041]	0.546	1.000	0.041	yes	no
P4	early-late	+0.010	[-0.018, +0.038]	0.554	1.000	0.038	yes	no
P4	early-gated	-0.006	[-0.042, +0.030]	0.781	1.000	0.042	yes	no
P4	early-attention	+0.018	[-0.010, +0.046]	0.277	1.000	0.046	yes	no
P4	early-score	+0.029	[-0.007, +0.065]	0.182	1.000	0.065	no	no
P4	late-gated	-0.016	[-0.035, +0.003]	0.160	1.000	0.035	yes	no
P4	late-attention	+0.008	[-0.005, +0.022]	0.305	1.000	0.022	yes	no
P4	late-score	+0.019	[-0.013, +0.051]	0.316	1.000	0.051	no	no
P4	gated-attention	+0.024	[+0.008, +0.041]	0.019	0.192	0.041	yes	no
P4	gated-score	+0.035	[+0.006, +0.064]	0.052	0.469	0.064	no	no
P4	attention-score	+0.011	[-0.013, +0.035]	0.456	1.000	0.035	yes	no
P5	early-late	-0.006	[-0.048, +0.037]	0.817	1.000	0.048	yes	no
P5	early-gated	-0.012	[-0.056, +0.032]	0.640	1.000	0.056	no	no
P5	early-attention	-0.015	[-0.062, +0.032]	0.595	1.000	0.062	no	no

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P5	early-score	-0.003	[-0.045, +0.038]	0.897	1.000	0.045	yes	no
P5	late-gated	-0.006	[-0.028, +0.016]	0.623	1.000	0.028	yes	no
P5	late-attention	-0.009	[-0.027, +0.009]	0.398	1.000	0.027	yes	no
P5	late-score	+0.003	[-0.029, +0.035]	0.890	1.000	0.035	yes	no
P5	gated-attention	-0.003	[-0.027, +0.021]	0.849	1.000	0.027	yes	no
P5	gated-score	+0.009	[-0.025, +0.043]	0.655	1.000	0.043	yes	no
P5	attention-score	+0.012	[-0.018, +0.042]	0.511	1.000	0.042	yes	no
P6a	early-late	-0.008	[-0.041, +0.025]	0.680	1.000	0.041	yes	no
P6a	early-gated	-0.021	[-0.052, +0.010]	0.254	1.000	0.052	no	no
P6a	early-attention	-0.006	[-0.040, +0.029]	0.775	1.000	0.040	yes	no
P6a	early-score	-0.003	[-0.032, +0.025]	0.856	1.000	0.032	yes	no
P6a	late-gated	-0.013	[-0.025, -0.001]	0.077	0.768	0.025	yes	no
P6a	late-attention	+0.002	[-0.017, +0.021]	0.851	1.000	0.021	yes	no
P6a	late-score	+0.005	[-0.025, +0.035]	0.782	1.000	0.035	yes	no
P6a	gated-attention	+0.015	[-0.004, +0.034]	0.191	1.000	0.034	yes	no
P6a	gated-score	+0.018	[-0.005, +0.041]	0.194	1.000	0.041	yes	no
P6a	attention-score	+0.003	[-0.027, +0.032]	0.875	1.000	0.032	yes	no
P6b	early-late	-0.025	[-0.064, +0.015]	0.291	1.000	0.064	no	no
P6b	early-gated	-0.029	[-0.066, +0.008]	0.187	1.000	0.066	no	no
P6b	early-attention	-0.019	[-0.059, +0.020]	0.403	1.000	0.059	no	no
P6b	early-score	-0.020	[-0.054, +0.014]	0.314	1.000	0.054	no	no
P6b	late-gated	-0.004	[-0.026, +0.017]	0.726	1.000	0.026	yes	no
P6b	late-attention	+0.005	[-0.021, +0.032]	0.730	1.000	0.032	yes	no
P6b	late-score	+0.005	[-0.030, +0.039]	0.826	1.000	0.039	yes	no
P6b	gated-attention	+0.010	[-0.017, +0.036]	0.531	1.000	0.036	yes	no
P6b	gated-score	+0.009	[-0.028, +0.046]	0.683	1.000	0.046	yes	no
P6b	attention-score	-0.001	[-0.030, +0.028]	0.960	1.000	0.030	yes	no
P6c	early-late	-0.017	[-0.063, +0.029]	0.532	1.000	0.063	no	no
P6c	early-gated	-0.017	[-0.051, +0.016]	0.378	1.000	0.051	no	no
P6c	early-attention	-0.008	[-0.042, +0.025]	0.676	1.000	0.042	yes	no
P6c	early-score	-0.003	[-0.035, +0.029]	0.872	1.000	0.035	yes	no
P6c	late-gated	-0.000	[-0.021, +0.020]	0.976	1.000	0.021	yes	no
P6c	late-attention	+0.009	[-0.012, +0.029]	0.470	1.000	0.029	yes	no
P6c	late-score	+0.014	[-0.026, +0.054]	0.555	1.000	0.054	no	no
P6c	gated-attention	+0.009	[-0.008, +0.026]	0.374	1.000	0.026	yes	no
P6c	gated-score	+0.014	[-0.019, +0.048]	0.469	1.000	0.048	yes	no
P6c	attention-score	+0.005	[-0.026, +0.037]	0.776	1.000	0.037	yes	no

Siena (difference: first minus second operator, macro F1; fold values averaged over the seed sets)

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P0	early-late	-0.015	[-0.056, +0.025]	0.513	1.000	0.056	no	no
P0	early-gated	-0.007	[-0.062, +0.048]	0.820	1.000	0.062	no	no
P0	early-attention	+0.003	[-0.048, +0.055]	0.910	1.000	0.055	no	no
P0	early-score	-0.038	[-0.081, +0.004]	0.132	1.000	0.081	no	no
P0	late-gated	+0.008	[-0.025, +0.041]	0.666	1.000	0.041	yes	no
P0	late-attention	+0.019	[-0.016, +0.053]	0.355	1.000	0.053	no	no
P0	late-score	-0.023	[-0.058, +0.012]	0.269	1.000	0.058	no	no
P0	gated-attention	+0.011	[-0.007, +0.028]	0.316	1.000	0.028	yes	no
P0	gated-score	-0.031	[-0.054, -0.009]	0.029	0.257	0.054	no	no
P0	attention-score	-0.042	[-0.066, -0.017]	0.009	0.095	0.066	no	no
P1	early-late	-0.020	[-0.095, +0.055]	0.643	1.000	0.095	no	no
P1	early-gated	-0.007	[-0.104, +0.091]	0.905	1.000	0.104	no	no
P1	early-attention	-0.012	[-0.106, +0.083]	0.833	1.000	0.106	no	no
P1	early-score	-0.049	[-0.106, +0.008]	0.149	1.000	0.106	no	no
P1	late-gated	+0.013	[-0.029, +0.056]	0.580	1.000	0.056	no	no
P1	late-attention	+0.009	[-0.028, +0.045]	0.685	1.000	0.045	yes	no
P1	late-score	-0.029	[-0.083, +0.026]	0.367	1.000	0.083	no	no

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P1	gated-attention	-0.005	[-0.039, +0.030]	0.806	1.000	0.039	yes	no
P1	gated-score	-0.042	[-0.110, +0.026]	0.291	1.000	0.110	no	no
P1	attention-score	-0.037	[-0.098, +0.024]	0.297	1.000	0.098	no	no
P2	early-late	-0.029	[-0.100, +0.042]	0.485	1.000	0.100	no	no
P2	early-gated	-0.026	[-0.107, +0.055]	0.581	1.000	0.107	no	no
P2	early-attention	-0.015	[-0.100, +0.071]	0.766	1.000	0.100	no	no
P2	early-score	-0.064	[-0.112, -0.016]	0.034	0.339	0.112	no	no
P2	late-gated	+0.003	[-0.023, +0.029]	0.844	1.000	0.029	yes	no
P2	late-attention	+0.014	[-0.013, +0.042]	0.379	1.000	0.042	yes	no
P2	late-score	-0.035	[-0.097, +0.026]	0.327	1.000	0.097	no	no
P2	gated-attention	+0.011	[-0.017, +0.040]	0.494	1.000	0.040	yes	no
P2	gated-score	-0.038	[-0.106, +0.029]	0.335	1.000	0.106	no	no
P2	attention-score	-0.050	[-0.127, +0.027]	0.274	1.000	0.127	no	no
P3	early-late	-0.038	[-0.110, +0.034]	0.362	1.000	0.110	no	no
P3	early-gated	-0.007	[-0.115, +0.102]	0.917	1.000	0.115	no	no
P3	early-attention	+0.000	[-0.093, +0.094]	0.994	1.000	0.094	no	no
P3	early-score	-0.050	[-0.127, +0.026]	0.261	1.000	0.127	no	no
P3	late-gated	+0.032	[-0.024, +0.088]	0.331	1.000	0.088	no	no
P3	late-attention	+0.039	[-0.011, +0.088]	0.189	1.000	0.088	no	no
P3	late-score	-0.012	[-0.042, +0.018]	0.483	1.000	0.042	yes	no
P3	gated-attention	+0.007	[-0.035, +0.048]	0.773	1.000	0.048	yes	no
P3	gated-score	-0.044	[-0.103, +0.015]	0.207	1.000	0.103	no	no
P3	attention-score	-0.051	[-0.100, -0.002]	0.089	0.893	0.100	no	no
P4	early-late	-0.022	[-0.096, +0.052]	0.607	1.000	0.096	no	no
P4	early-gated	-0.007	[-0.101, +0.087]	0.901	1.000	0.101	no	no
P4	early-attention	-0.012	[-0.106, +0.081]	0.817	1.000	0.106	no	no
P4	early-score	-0.051	[-0.108, +0.005]	0.130	1.000	0.108	no	no
P4	late-gated	+0.015	[-0.026, +0.056]	0.521	1.000	0.056	no	no
P4	late-attention	+0.010	[-0.027, +0.046]	0.650	1.000	0.046	yes	no
P4	late-score	-0.029	[-0.082, +0.023]	0.341	1.000	0.082	no	no
P4	gated-attention	-0.006	[-0.040, +0.028]	0.771	1.000	0.040	yes	no
P4	gated-score	-0.045	[-0.111, +0.022]	0.254	1.000	0.111	no	no
P4	attention-score	-0.039	[-0.099, +0.021]	0.269	1.000	0.099	no	no
P5	early-late	-0.005	[-0.071, +0.061]	0.895	1.000	0.071	no	no
P5	early-gated	-0.013	[-0.086, +0.061]	0.767	1.000	0.086	no	no
P5	early-attention	-0.015	[-0.080, +0.051]	0.700	1.000	0.080	no	no
P5	early-score	-0.053	[-0.112, +0.007]	0.142	1.000	0.112	no	no
P5	late-gated	-0.008	[-0.040, +0.025]	0.691	1.000	0.040	yes	no
P5	late-attention	-0.010	[-0.058, +0.038]	0.727	1.000	0.058	no	no
P5	late-score	-0.048	[-0.097, +0.002]	0.111	1.000	0.097	no	no
P5	gated-attention	-0.002	[-0.045, +0.040]	0.931	1.000	0.045	yes	no
P5	gated-score	-0.040	[-0.084, +0.004]	0.132	1.000	0.084	no	no
P5	attention-score	-0.038	[-0.084, +0.008]	0.167	1.000	0.084	no	no
P6a	early-late	-0.003	[-0.074, +0.067]	0.933	1.000	0.074	no	no
P6a	early-gated	-0.002	[-0.049, +0.044]	0.936	1.000	0.049	yes	no
P6a	early-attention	+0.008	[-0.052, +0.067]	0.824	1.000	0.067	no	no
P6a	early-score	-0.053	[-0.113, +0.007]	0.139	1.000	0.113	no	no
P6a	late-gated	+0.001	[-0.044, +0.046]	0.960	1.000	0.046	yes	no
P6a	late-attention	+0.011	[-0.019, +0.041]	0.524	1.000	0.041	yes	no
P6a	late-score	-0.050	[-0.148, +0.048]	0.385	1.000	0.148	no	no
P6a	gated-attention	+0.010	[-0.025, +0.044]	0.625	1.000	0.044	yes	no
P6a	gated-score	-0.051	[-0.132, +0.030]	0.282	1.000	0.132	no	no
P6a	attention-score	-0.061	[-0.144, +0.022]	0.218	1.000	0.144	no	no
P6b	early-late	-0.067	[-0.110, -0.023]	0.018	0.178	0.110	no	no
P6b	early-gated	-0.070	[-0.124, -0.015]	0.042	0.338	0.124	no	no
P6b	early-attention	-0.038	[-0.077, +0.001]	0.106	0.745	0.077	no	no
P6b	early-score	-0.062	[-0.105, -0.019]	0.024	0.219	0.105	no	no
P6b	late-gated	-0.003	[-0.021, +0.015]	0.786	1.000	0.021	yes	no
P6b	late-attention	+0.029	[-0.006, +0.064]	0.170	1.000	0.064	no	no
P6b	late-score	+0.005	[-0.033, +0.043]	0.819	1.000	0.043	yes	no
P6b	gated-attention	+0.031	[-0.015, +0.078]	0.255	1.000	0.078	no	no

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P6b	gated-score	+0.008	[-0.040, +0.056]	0.779	1.000	0.056	no	no
P6b	attention-score	-0.024	[-0.062, +0.015]	0.293	1.000	0.062	no	no
P6c	early-late	-0.009	[-0.057, +0.038]	0.738	1.000	0.057	no	no
P6c	early-gated	+0.016	[-0.045, +0.077]	0.656	1.000	0.077	no	no
P6c	early-attention	+0.007	[-0.048, +0.063]	0.817	1.000	0.063	no	no
P6c	early-score	-0.054	[-0.114, +0.006]	0.133	1.000	0.114	no	no
P6c	late-gated	+0.025	[-0.016, +0.066]	0.304	1.000	0.066	no	no
P6c	late-attention	+0.017	[-0.010, +0.044]	0.297	1.000	0.044	yes	no
P6c	late-score	-0.045	[-0.111, +0.022]	0.254	1.000	0.111	no	no
P6c	gated-attention	-0.008	[-0.039, +0.022]	0.636	1.000	0.039	yes	no
P6c	gated-score	-0.070	[-0.146, +0.007]	0.131	1.000	0.146	no	no
P6c	attention-score	-0.061	[-0.129, +0.006]	0.131	1.000	0.129	no	no

Table S9: Post hoc: sensitivity to the decision threshold (window level). Exploratory, specified after the primary analysis (Section 2.6). Every model's stored test predictions were thresholded at 0.3, 0.5 and 0.7; 0.5 is the primary analysis (`python -m src.sensitivity_preds`, `src.posthoc_checks`; Figure 9a). **S9a.** Macro F1 per pipeline and threshold, mean over the five seed sets of each run's fold mean.
CHB-MIT

pipeline	threshold	early	late	gated	attention	score	spec2d_wide	shallow
P0	0.3	0.678	0.685	0.684	0.688	0.685	0.686	0.612
P0	0.5	0.678	0.679	0.671	0.683	0.661	0.680	0.671
P0	0.7	0.668	0.668	0.660	0.670	0.618	0.667	0.705
P1	0.3	0.687	0.676	0.699	0.699	0.691	0.694	0.645
P1	0.5	0.684	0.665	0.691	0.687	0.672	0.691	0.695
P1	0.7	0.674	0.658	0.679	0.672	0.632	0.672	0.719
P2	0.3	0.686	0.697	0.689	0.695	0.698	0.686	0.641
P2	0.5	0.680	0.688	0.680	0.687	0.675	0.682	0.687
P2	0.7	0.670	0.675	0.670	0.674	0.623	0.667	0.712
P3	0.3	0.680	0.684	0.684	0.698	0.689	0.689	0.652
P3	0.5	0.679	0.677	0.676	0.689	0.678	0.686	0.695
P3	0.7	0.673	0.668	0.667	0.673	0.634	0.678	0.723
P4	0.3	0.692	0.684	0.701	0.682	0.688	0.687	0.642
P4	0.5	0.688	0.678	0.694	0.670	0.660	0.679	0.692
P4	0.7	0.677	0.667	0.679	0.657	0.613	0.661	0.715
P5	0.3	0.677	0.689	0.697	0.697	0.701	0.690	0.645
P5	0.5	0.678	0.684	0.690	0.693	0.681	0.687	0.695
P5	0.7	0.673	0.672	0.678	0.685	0.644	0.668	0.719
P6a	0.3	0.677	0.686	0.703	0.689	0.691	0.683	0.644
P6a	0.5	0.672	0.679	0.692	0.677	0.675	0.685	0.687
P6a	0.7	0.664	0.670	0.676	0.664	0.630	0.665	0.710
P6b	0.3	0.661	0.694	0.693	0.685	0.682	0.674	0.642
P6b	0.5	0.662	0.686	0.691	0.681	0.682	0.670	0.702
P6b	0.7	0.654	0.673	0.681	0.665	0.637	0.655	0.724
P6c	0.3	0.680	0.694	0.696	0.688	0.690	0.686	0.652
P6c	0.5	0.671	0.688	0.688	0.679	0.674	0.689	0.688
P6c	0.7	0.661	0.676	0.677	0.665	0.638	0.673	0.714

Siena

pipeline	threshold	early	late	gated	attention	score	spec2d_wide	shallow
P0	0.3	0.590	0.594	0.561	0.574	0.574	0.591	0.656
P0	0.5	0.601	0.616	0.608	0.598	0.639	0.618	0.692
P0	0.7	0.600	0.622	0.599	0.603	0.642	0.631	0.692
P1	0.3	0.601	0.615	0.595	0.611	0.615	0.656	0.703
P1	0.5	0.616	0.637	0.623	0.628	0.665	0.668	0.729
P1	0.7	0.619	0.648	0.633	0.636	0.670	0.668	0.728

pipeline	threshold	early	late	gated	attention	score	spec2d_wide	shallow
P2	0.3	0.588	0.619	0.610	0.600	0.605	0.658	0.707
P2	0.5	0.610	0.639	0.636	0.625	0.674	0.674	0.747
P2	0.7	0.620	0.650	0.640	0.640	0.658	0.669	0.750
P3	0.3	0.581	0.617	0.582	0.575	0.609	0.643	0.688
P3	0.5	0.613	0.652	0.620	0.613	0.664	0.654	0.711
P3	0.7	0.608	0.653	0.630	0.629	0.667	0.655	0.725
P4	0.3	0.599	0.615	0.591	0.609	0.611	0.658	0.703
P4	0.5	0.614	0.637	0.621	0.627	0.666	0.671	0.729
P4	0.7	0.618	0.648	0.632	0.635	0.670	0.670	0.728
P5	0.3	0.609	0.629	0.630	0.617	0.627	0.657	0.703
P5	0.5	0.637	0.642	0.650	0.652	0.690	0.664	0.729
P5	0.7	0.636	0.652	0.650	0.653	0.669	0.669	0.728
P6a	0.3	0.604	0.596	0.578	0.582	0.629	0.669	0.671
P6a	0.5	0.619	0.622	0.621	0.611	0.672	0.677	0.702
P6a	0.7	0.619	0.617	0.624	0.607	0.649	0.657	0.705
P6b	0.3	0.614	0.671	0.661	0.623	0.654	0.652	0.657
P6b	0.5	0.614	0.681	0.684	0.652	0.676	0.665	0.699
P6b	0.7	0.612	0.669	0.676	0.647	0.645	0.673	0.716
P6c	0.3	0.604	0.613	0.567	0.589	0.645	0.642	0.695
P6c	0.5	0.629	0.638	0.613	0.621	0.683	0.673	0.746
P6c	0.7	0.625	0.636	0.620	0.612	0.662	0.672	0.747

S9b. Each operator's mean rank (1 = best) over the 45 runs of a dataset and the share of runs in which it ranked first (ties share the rank), and the share of runs whose best operator differs from the best at 0.5.

dataset	threshold	early: mean rank / first	late: mean rank / first	gated: mean rank / first	attention: mean rank / first	score: mean rank / first	winner differs from 0.5
CHB-MIT	0.3	3.76 / 7%	3.18 / 24%	2.69 / 36%	2.73 / 20%	2.64 / 13%	27%
CHB-MIT	0.5	3.20 / 11%	2.93 / 20%	2.64 / 38%	2.64 / 24%	3.58 / 7%	0%
CHB-MIT	0.7	2.62 / 22%	2.69 / 16%	2.27 / 40%	2.49 / 22%	4.93 / 0%	33%
Siena	0.3	3.42 / 13%	2.42 / 27%	3.40 / 11%	3.31 / 11%	2.44 / 38%	53%
Siena	0.5	3.93 / 0%	2.84 / 7%	3.24 / 9%	3.69 / 2%	1.29 / 82%	0%
Siena	0.7	4.00 / 2%	2.53 / 11%	3.31 / 9%	3.51 / 11%	1.64 / 67%	29%

S9c. Operator pairs with a raw p below 0.05 at any threshold: corrected resampled t test on the fold values (seed sets averaged), Holm-adjusted within the ten pairs of one dataset, pipeline and threshold (the family of the primary analysis), and across the thirty comparisons of one dataset and pipeline (three thresholds x ten pairs), a more conservative sensitivity check. Difference: first minus second operator.

dataset	pipeline	threshold	pair	difference	raw p	Holm within threshold	Holm across thresholds
CHB-MIT	P4	0.5	gated-attention	+0.024	0.0192	0.1922	0.4997
CHB-MIT	P0	0.7	late-score	+0.050	0.0203	0.2034	0.6102
CHB-MIT	P0	0.7	attention-score	+0.052	0.0337	0.3031	0.9766
CHB-MIT	P2	0.7	late-score	+0.052	0.0118	0.1179	0.3537
CHB-MIT	P2	0.7	gated-score	+0.047	0.0233	0.2101	0.6770
CHB-MIT	P2	0.7	attention-score	+0.051	0.0390	0.3119	1.0000
CHB-MIT	P4	0.7	early-score	+0.064	0.0227	0.1363	0.5679
CHB-MIT	P4	0.7	late-score	+0.054	0.0155	0.1237	0.4329
CHB-MIT	P4	0.7	gated-attention	+0.021	0.0038	0.0344	0.1109
CHB-MIT	P4	0.7	gated-score	+0.066	0.0019	0.0194	0.0583
CHB-MIT	P4	0.7	attention-score	+0.044	0.0155	0.1237	0.4329
CHB-MIT	P6a	0.7	gated-score	+0.046	0.0347	0.3469	1.0000
CHB-MIT	P6b	0.7	late-score	+0.036	0.0369	0.3688	1.0000

dataset	pipeline	threshold	pair	difference	raw p	Holm within threshold	Holm across thresholds
Siena	P6b	0.3	early-late	-0.057	0.0316	0.2846	0.8144
Siena	P6b	0.3	late-attention	+0.048	0.0255	0.2550	0.6885
Siena	P0	0.5	gated-score	-0.031	0.0285	0.2568	0.8273
Siena	P0	0.5	attention-score	-0.042	0.0095	0.0945	0.2836
Siena	P2	0.5	early-score	-0.064	0.0339	0.3393	1.0000
Siena	P6b	0.5	early-late	-0.067	0.0178	0.1783	0.5172
Siena	P6b	0.5	early-gated	-0.070	0.0423	0.3382	1.0000
Siena	P6b	0.5	early-score	-0.062	0.0243	0.2191	0.6816
Siena	P3	0.7	early-score	-0.059	0.0199	0.1990	0.5971
Siena	P6b	0.7	early-late	-0.057	0.0148	0.1482	0.4446
Siena	P6b	0.7	early-gated	-0.064	0.0313	0.2819	0.8144

All 540 comparisons (2 datasets x 9 pipelines x 3 thresholds x 10 pairs) are in `results_v2/sensitivity_preds/posthoc_holm_thresholds.csv`; 2 are significant after the Holm adjustment within a threshold and 0 after the adjustment across thresholds.

Table S10: Post hoc: seizure-level thresholds, class prevalence and inference. Exploratory, specified after the primary analysis (Section 2.6). **S10a. Seizure-level metrics at three thresholds**, P0 and P1 only: every recording of every held-out person scored again with the saved models (`python -m src.events --thresholds 0.3 0.5 0.7`), no post-processing; pooled over the persons of a run, mean ± SD over the five seed sets. False alarms per hour are given for CHB-MIT only (Siena's recordings are cut around seizures). **CHB-MIT**

pipeline	model	sensitivity 0.3	sensitivity 0.5	sensitivity 0.7	false alarms/h 0.3	false alarms/h 0.5	false alarms/h 0.7
P0	early	0.83 ±0.03	0.79 ±0.04	0.70 ±0.05	8.87 ±0.97	7.75 ±0.79	6.73 ±0.71
P0	late	0.74 ±0.06	0.69 ±0.06	0.64 ±0.06	6.39 ±0.55	5.50 ±0.50	4.66 ±0.51
P0	gated	0.73 ±0.05	0.67 ±0.05	0.61 ±0.04	6.18 ±0.53	5.32 ±0.47	4.48 ±0.51
P0	attention	0.73 ±0.03	0.68 ±0.03	0.63 ±0.03	6.34 ±0.60	5.38 ±0.60	4.50 ±0.60
P0	score	0.80 ±0.04	0.65 ±0.05	0.48 ±0.06	8.15 ±0.25	5.40 ±0.45	3.17 ±0.33
P0	spec2d_wide	0.87 ±0.03	0.79 ±0.04	0.73 ±0.04	7.76 ±0.28	6.40 ±0.44	5.00 ±0.47
P0	shallow	0.86 ±0.00	0.82 ±0.00	0.76 ±0.00	9.46 ±0.00	7.93 ±0.00	6.22 ±0.00
P1	early	0.82 ±0.04	0.77 ±0.04	0.71 ±0.03	8.29 ±0.43	7.01 ±0.28	5.82 ±0.28
P1	late	0.73 ±0.07	0.68 ±0.06	0.63 ±0.05	6.17 ±0.49	5.31 ±0.62	4.56 ±0.72
P1	gated	0.76 ±0.05	0.71 ±0.05	0.65 ±0.04	6.04 ±0.55	5.16 ±0.60	4.36 ±0.65
P1	attention	0.80 ±0.04	0.73 ±0.04	0.66 ±0.03	6.62 ±0.66	5.57 ±0.64	4.68 ±0.61
P1	score	0.82 ±0.02	0.67 ±0.04	0.53 ±0.04	8.46 ±0.32	5.90 ±0.37	3.35 ±0.36
P1	spec2d_wide	0.87 ±0.03	0.81 ±0.03	0.75 ±0.04	7.99 ±0.65	6.67 ±0.46	5.39 ±0.32
P1	shallow	0.90 ±0.00	0.86 ±0.00	0.79 ±0.00	9.70 ±0.00	8.25 ±0.00	6.47 ±0.00

Siena

pipeline	model	sensitivity 0.3	sensitivity 0.5	sensitivity 0.7
P0	early	0.94 ±0.05	0.91 ±0.02	0.85 ±0.05
P0	late	0.86 ±0.06	0.83 ±0.08	0.77 ±0.09
P0	gated	0.90 ±0.04	0.85 ±0.06	0.78 ±0.09
P0	attention	0.92 ±0.06	0.87 ±0.04	0.79 ±0.05
P0	score	0.99 ±0.01	0.92 ±0.05	0.74 ±0.07
P0	spec2d_wide	0.89 ±0.03	0.85 ±0.02	0.77 ±0.05
P0	shallow	0.98 ±0.00	0.96 ±0.00	0.87 ±0.00
P1	early	0.86 ±0.06	0.82 ±0.07	0.78 ±0.06
P1	late	0.88 ±0.02	0.86 ±0.03	0.81 ±0.03
P1	gated	0.86 ±0.07	0.86 ±0.07	0.80 ±0.08
P1	attention	0.89 ±0.11	0.86 ±0.12	0.81 ±0.10
P1	score	0.94 ±0.03	0.84 ±0.04	0.71 ±0.07

pipeline	model	sensitivity 0.3	sensitivity 0.5	sensitivity 0.7
P1	spec2d_wide	0.91 ±0.06	0.87 ±0.05	0.83 ±0.05
P1	shallow	1.00 ±0.00	0.98 ±0.00	0.94 ±0.00

S10b. CHB-MIT prevalence-adjusted macro F1. Each non-ictal test window is weighted by the number of non-ictal windows of its case in the complete recordings divided by the number sampled from that case (ictal windows: weight 1), so true positives and false negatives keep their counts and false positives and true negatives are scaled to the full non-ictal count; macro F1 is computed from this weighted confusion matrix at the 0.5 threshold. Subject-level: macro F1 per held-out person, mean over the 23 persons, then mean and SD over the 45 runs; pooled: the weighted confusion matrices of a run's persons summed first. False positive rate: share of the sampled non-ictal test windows predicted ictal at 0.5 (unchanged by the weighting); precision at 0.5 (adjusted): TP / (TP + weighted FP). The estimate assumes that the sampled non-ictal windows have the error rate of those not sampled; it is not a measurement on all windows (`src/posthoc_checks.py`).

aggregation	model	sampled	sampled SD	adjusted	adjusted SD	decline	relative decline	false positive rate at 0.5	precision at 0.5 (adjusted)
subject-level	early	0.677	0.013	0.517	0.016	0.160	23.7%	0.110	0.113
subject-level	late	0.681	0.015	0.555	0.015	0.126	18.5%	0.078	0.226
subject-level	gated	0.686	0.016	0.562	0.015	0.124	18.1%	0.074	0.244
subject-level	attention	0.683	0.014	0.558	0.014	0.125	18.3%	0.082	0.223
subject-level	score	0.673	0.017	0.545	0.014	0.128	18.9%	0.089	0.194
subject-level	spec2d_wide	0.683	0.014	0.541	0.014	0.142	20.8%	0.088	0.162
subject-level	shallow	0.690	0.008	0.454	0.004	0.237	34.3%	0.215	0.022
pooled	early	0.678	0.014	0.483	0.007	0.195	28.7%	0.103	0.014
pooled	late	0.694	0.020	0.498	0.008	0.196	28.2%	0.085	0.020
pooled	gated	0.698	0.021	0.500	0.008	0.198	28.3%	0.079	0.022
pooled	attention	0.701	0.018	0.496	0.008	0.205	29.3%	0.083	0.020
pooled	score	0.690	0.018	0.492	0.008	0.198	28.7%	0.089	0.018
pooled	spec2d_wide	0.684	0.014	0.489	0.006	0.195	28.5%	0.083	0.016
pooled	shallow	0.671	0.009	0.447	0.003	0.224	33.4%	0.183	0.009

Weights per case (all windows: every 10 s window of the case's target-montage recordings):

case	all windows	ictal	non-ictal, full	non-ictal, sampled	weight w
chb01	14598	50	14548	200	72.7
chb02	12695	20	12675	80	158.4
chb03	13680	45	13635	180	75.8
chb04	56176	42	56134	168	334.1
chb05	14041	62	13979	248	56.4
chb06	24023	24	23999	96	250.0
chb07	24138	36	24102	144	167.4
chb08	7202	97	7105	388	18.3
chb09	24430	31	24399	124	196.8
chb10	18007	49	17958	196	91.6
chb11	12524	83	12441	332	37.5
chb12	7446	123	7323	492	14.9
chb13	11880	67	11813	268	44.1
chb14	9360	24	9336	96	97.2
chb15	14402	214	14188	856	16.6

case	all windows	ictal	non-ictal, full	non-ictal, sampled	weight w
chb16	6840	14	6826	56	121.9
chb17	7562	32	7530	128	58.8
chb18	12828	38	12790	152	84.1
chb19	10774	27	10747	108	99.5
chb20	9933	36	9897	144	68.7
chb21	11818	24	11794	96	122.9
chb22	11161	24	11137	96	116.0
chb23	9558	49	9509	196	48.5
chb24	7666	65	7601	260	29.2

S10c. Corrected paired t test and subject-level bootstrap, operator pairs pooled over the nine pipelines (each subject's difference averaged over pipelines and seed sets). Corrected 95% interval: mean ± t(0.975, n-1) SE with the Nadeau-Bengio variance. Bootstrap: the held-out subject is the resampling unit, and both operators' results of a drawn subject enter together (paired); 10,000 resamples, percentile intervals; the second interval also resamples the five seed sets within each drawn subject. Direction consistency: share of resampled means with the sign of the observed difference. The bootstrap does not model the overlap of the LOSO training sets; the corrected test remains the primary analysis. Assumptions: the corrected test treats the per-subject differences as approximately normal and approximates the dependence between folds by the Nadeau-Bengio variance inflation (test/train ratio 1/22 and 1/13); the percentile bootstrap treats the held-out subjects as independent draws from a population of subjects, and with 23 and 14 subjects its intervals may be too narrow. Neither interval is adjusted for multiplicity: they are comparable with each other, whereas the primary analysis applies Holm adjustment within each pipeline. All 90 pairs per dataset in `results_v2/sensitivity_preds/posthoc_bootstrap_compare.csv`.

dataset	pair	mean difference	corrected 95% CI	corrected p	bootstrap 95% CI (subject)	bootstrap 95% CI (subject, then seed set)	direction consistency
CHB-MIT	early-late	-0.004	[-0.041, +0.034]	0.8333	[-0.028, +0.020]	[-0.029, +0.020]	62.0%
CHB-MIT	early-gated	-0.009	[-0.045, +0.027]	0.5982	[-0.033, +0.013]	[-0.033, +0.013]	78.9%
CHB-MIT	early-attention	-0.006	[-0.045, +0.032]	0.7440	[-0.031, +0.018]	[-0.031, +0.019]	69.0%
CHB-MIT	early-score	+0.004	[-0.030, +0.037]	0.8174	[-0.018, +0.025]	[-0.018, +0.025]	63.3%
CHB-MIT	late-gated	-0.005	[-0.016, +0.005]	0.2921	[-0.012, +0.001]	[-0.014, +0.003]	95.0%
CHB-MIT	late-attention	-0.002	[-0.014, +0.010]	0.6971	[-0.010, +0.005]	[-0.012, +0.008]	70.8%
CHB-MIT	late-score	+0.008	[-0.022, +0.037]	0.6007	[-0.011, +0.028]	[-0.011, +0.028]	77.7%
CHB-MIT	gated-attention	+0.003	[-0.011, +0.017]	0.6534	[-0.006, +0.012]	[-0.007, +0.014]	74.5%
CHB-MIT	gated-score	+0.013	[-0.014, +0.040]	0.3280	[-0.003, +0.032]	[-0.004, +0.033]	93.6%
CHB-MIT	attention-score	+0.010	[-0.017, +0.037]	0.4515	[-0.007, +0.028]	[-0.008, +0.029]	86.9%
Siena	early-late	-0.023	[-0.090, +0.043]	0.4640	[-0.059, +0.021]	[-0.060, +0.022]	86.5%
Siena	early-gated	-0.014	[-0.092, +0.065]	0.7171	[-0.057, +0.038]	[-0.058, +0.039]	71.9%
Siena	early-attention	-0.008	[-0.086, +0.070]	0.8276	[-0.050, +0.044]	[-0.051, +0.045]	64.8%
Siena	early-score	-0.053	[-0.101, -0.005]	0.0337	[-0.081, -0.024]	[-0.083, -0.022]	100.0%
Siena	late-gated	+0.010	[-0.012, +0.031]	0.3483	[-0.002, +0.024]	[-0.005, +0.027]	93.8%
Siena	late-attention	+0.015	[-0.010, +0.040]	0.2145	[+0.002, +0.033]	[-0.000, +0.034]	99.4%
Siena	late-score	-0.030	[-0.078, +0.018]	0.2065	[-0.060, -0.003]	[-0.062, -0.002]	98.7%
Siena	gated-attention	+0.005	[-0.014, +0.025]	0.5513	[-0.007, +0.016]	[-0.011, +0.020]	82.2%
Siena	gated-score	-0.039	[-0.092, +0.014]	0.1323	[-0.073, -0.010]	[-0.075, -0.008]	99.9%
Siena	attention-score	-0.045	[-0.095, +0.006]	0.0768	[-0.079, -0.019]	[-0.080, -0.017]	100.0%

Pairs per pipeline (90 per dataset) whose 95% interval excludes zero; only the Holm column is adjusted for multiplicity:

dataset	corrected test, nominal (raw p < 0.05)	corrected test, Holm-adjusted p < 0.05	bootstrap (subject), nominal	bootstrap (subject, then seed set), nominal
CHB-MIT	1/90	0/90	10/90	3/90
Siena	6/90	0/90	24/90	17/90

S10d. CHB-MIT prevalence-adjusted operator comparisons. The procedure of the primary analysis applied to the prevalence-adjusted fold values of S10b: per held-out subject the fold value of each operator, seed sets averaged; corrected resampled t test (Nadeau-Bengio, test/train ratio 1/22, 22 degrees of freedom) on the 23 paired differences; Holm adjustment within the ten operator pairs of one pipeline (nine families per metric). Difference: first minus second operator. Post hoc and exploratory (`results_v2/sensitivity_preds/posthoc_prevalence_pairs.csv`).

metric	raw p < 0.05	Holm-adjusted p < 0.05
AUPRC (adjusted)	5/90	0/90
macro F1 (adjusted)	22/90	3/90

Holm-significant comparisons:

metric	pipeline	pair	mean difference	corrected 95% CI	t	raw p	Holm-adjusted p
macro F1 (adjusted)	P3	early-late	-0.042	[-0.069, -0.014]	-3.17	0.0044	0.0399
macro F1 (adjusted)	P3	early-attention	-0.057	[-0.091, -0.023]	-3.48	0.0021	0.0214
macro F1 (adjusted)	P5	early-gated	-0.068	[-0.106, -0.029]	-3.66	0.0014	0.0138

Table S11: Reproducibility map. Where each analysis of the manuscript is implemented and where its results are stored in the public repository. Every `src/...` module runs as `python -m src.<module>`; the README lists the full commands and their order.

Manuscript	Code	Results and records
2.1 CHB-MIT annotations (seizures assigned by file name; `chb09_08.edf`), counts of recordings and seizures	`src/chbmit.py` (`parse_summary`), `src/verify_chbmit.py`, `src/audit_methods.py`	`results_v2/audit/methods_audit.md`, `counts.csv`
2.1 Siena seizure lists, the six decisions, resampling to 256 Hz and the bipolar montage	`src/siena.py` (`read_bipolar`), `src/siena_qc.py`, `src/verify_siena.py`; `docs/SIENA.md`	Table S4
2.2 Preprocessing pipelines P0 to P6c; spectrogram; GEDAI preset (Table S2)	`src/preprocess.py`, `src/chbmit_prep.py`, `src/gedai_qc.py`	per run `meta.json`, `results_v2/qc/gedai_settings/`

Manuscript	Code	Results and records
2.2 Artefact removal records per recording	src/preprocess.py	results_v2/qc/p6_recording_logs.csv, p6_recording_logs_events.csv
2.3 Models, parameter matching	src/models.py (python -m src.models prints the counts)	
2.4 Windows, LOSO partitions, inner validation, leakage check and its negative control	src/chbmit_corpus.py, src/datasets.py, src/audit_methods.py	results_v2/audit/methods_audit.md, partitions_<ds>.csv
2.4 Seeds per seed set, fold and model	src/config.py (fold_seed)	per run meta.json
2.4, 3.7 Training on CHB-MIT, testing on Siena	src/cross_dataset.py	results_v2/cross/, results_v2/results/cross_*.csv
2.5, 3.6 Seizure-level scoring (rules fixed before the runs), the 187 reference events	src/events.py; docs/EVENTS.md; src/audit_methods.py	per run events/, results_v2/results/events_<ds>.csv
2.6 Simulation of the naive paired test	src/sim_cv_correlation.py	
2.6, 3.1 to 3.4 Corrected tests, equivalence, rankings, RM-ANOVA, variance components, conservative test/train ratio	src/multiverse.py, src/stats.py, src/audit_methods.py	results_v2/multiverse/chbmit_cuda/, siena_cuda/, results_v2/audit/methods_audit.md
3.1 Controls against the operators	src/controls_stats.py	results_v2/results/controls_*.csv, controls_tests.md
3.1 to 3.7 Tables and figures	src/make_results.py, src/make_figures.py, src/make_figure1.py, src/make_supplementary.py	results_v2/results/, paper/figures_manuscript/
2.6, 3.8 Post hoc threshold, prevalence and bootstrap analyses (Tables S9, S10)	src/sensitivity_preds.py, src/posthoc_checks.py; seizure-level thresholds: src/events.py --thresholds	results_v2/sensitivity_preds/
2.7 CPU against GPU, repeated runs	src/compare_runs.py, src/audit_methods.py; docs/PIPELINE.md	results_v2/audit/methods_audit.md
2.7, 3.5 Reproducibility of artefact removal across machines, perturbation test, impact on the results, signal effects (Tables S1, S5)	src/ica_repro.py, src/ica_sensitivity.py, src/ica_repro_impact.py, src/p6_signal_effects.py, src/gedai_qc.py; docs/ICA_REPRO.md	results_v2/ica_repro/, results_v2/qc/

Supplementary Note 1. The Bonn corpus (earlier stage)

An earlier stage of this project compared the same operators and controls on the Bonn corpus (Andrzejak et al. 2001): five sets of 100 single-channel segments (23.6 s, 173.61 Hz); A and B scalp recordings of healthy volunteers, C and D interictal and E ictal intracranial recordings. Three label groupings were used: T1, three classes (A,B / C,D / E); T2, intracranial only (C,D

vs E); T3, the common binary split (A,B,C,D vs E). Evaluation was 5 x 5 repeated stratified cross-validation over segments: Bonn has no subject identifiers, so a subject-wise split, as in the main study, is not possible. One preprocessing pipeline, CPU, earlier code version; these results are not part of the multiverse.

Model	T1 F1	T1 AUC	T2 F1	T2 AUC	T3 F1	T3 AUC	Params
late	0.974	0.999	0.987	1.000	0.986	1.000	98,571
gated	0.973	0.999	0.987	0.999	0.980	1.000	98,698
attention	0.969	0.997	0.985	0.999	0.977	0.998	98,544
spec2d_wide	0.966	0.998	0.984	0.999	0.982	1.000	97,775
score	0.964	0.998	0.981	0.999	0.976	0.999	59,302
raw1d_wide	0.963	0.997	0.973	0.998	0.976	0.992	99,795
spec2d	0.960	0.998	0.979	0.999	0.972	0.999	32,227
shallow	0.954	0.996	0.964	0.995	0.964	0.995	8
raw1d	0.951	0.996	0.964	0.997	0.969	0.995	27,075
early	0.896	0.977	0.923	0.981	0.899	0.965	97,507
logvar	0.646	0.755	0.933	0.986	0.954	0.992	1

Macro F1 and AUC, mean over the 25 paired measurements (`results_v2/T{1,2,3}_*/perfold.csv`; parameter counts of that code version, single-channel input).

Findings: (1) the naive paired Wilcoxon test found 22, 20 and 13 of 55 pairs significant on T1, T2 and T3; the corrected test found 10, 0 and 0, all ten on T1 involving logvar and none between two deep models (`results_v2/phase0/summary.csv`); (2) late, gated, attention and score fusion were equivalent to each other within about 0.02 to 0.03 macro F1 on T1; (3) early fusion was the one operator that separated, and only downward; (4) T3 is close to saturation: a single-feature logistic regression (logvar) reached 0.954 macro F1 against 0.986 for the best model.